Costly Signaling with Heterogeneous Outside Options

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Abstract

This paper studies optimal mechanism design in a setting with costly signaling and heterogeneous outside options. We consider a continuum model of continuous agent types and discrete common object qualities, where a planner wishes to maximize social welfare. An agent type consists of her value for quality and an independently drawn heterogeneous outside option. Our first main result considers a setting where heterogeneous outside options are observed by the planner. We show that the optimal incentive-compatible and individually rational mechanism determines a participation threshold for each outside option; an agent with a type below that threshold is assigned to her outside option, whereas an agent with type above that threshold is offered either a single common object quality or a lottery of consecutive common object qualities – potentially pooled with her outside option. Our second result shows that the same mechanism is also optimal in a setting with downward-private outside options, where agents can misreport having a lower outside option than their true one. Our framework highlights the important role of outside options in the design of ubiquitous resource allocation systems such as public or affordable housing, vaccine distribution and organ transplantation.

1 Introduction

In various contexts a social planner is faced with the decision of how to ration public resources in settings with costly signaling. In the absence of money, such goods are often rationed using priority-based wait-lists and other systems that may require costly signaling from agents. This paper is concerned with how the presence of heterogeneous (and possibly privately known) outside options impacts this social welfare maximization problem. For example, priorities in the allocation of a test or a vaccine during a pandemic flu may depend on the ability to work from home or where and with whom the person lives (Emanuel et al., 2020; Schmidt et al., 2020). Public housing authorities may screen families not only by their taste for available developments but also by their need or existing living conditions (Kaplan, 1988; Arnosti and Shi, 2020). The allocation of organs for transplantation may depend on the patients' access to other treatments, as well as their health condition and their sensitivity to match (Agarwal et al., 2021).[1]

We consider a setting in which resources of different qualities are to be allocated.[2] Agents have heterogeneous taste for quality and can signal their preference by taking costly actions. In addition, agents have different type-independent outside options. In particular, we consider a continuum of agents where each agent is characterized by a privately known type and a heterogeneous outside option, drawn independently from her type. We assume that the value for receiving an object is the product of her type and the object quality (thus the type can be interpreted as the willingness to pay for quality). The agent payoff is quasi-linear in the cost of her action.

We assume that the agents' outside options are *downward-private*, meaning that agents can *deflate* their outside option (i.e., they may report a "worse" outside option than their real one) but cannot inflate it. The rationale for the latter assumption is that, in several contexts, authorities can ask for evidence supporting the information reported by an individual. (For example, public housing authorities can ask for evidence regarding income or alternative housing options.) Then, while individuals may withhold information about their available options (e.g., the possibility of living at a family house), they may not be able to provide evidence for out-of-reach options.

The main result is a constructive characterization of the mechanism that maximizes social welfare, defined as the agents' average payoff. The optimal incentive-compatible and individually rational mechanism determines a participation threshold for each outside option; an agent with a type below that threshold is assigned to her outside option, while an agent with a type above that threshold is offered either a single object quality or a lottery of consecutive object qualities – potentially pooled with her outside option.

Our analysis first addresses the simpler case in which heterogeneous outside options are publicly known and therefore observed by the planner. The key challenge arises from the coexistence of

[1] The ethical principles for allocation of human organs state that factors to be considered include patient survival, graft survival, quality of life, age, and availability of alternative treatments. See https://optn.transplant.hrsa.gov/resources/ethics/ethical-principles-in-the-allocation-of-human-organs/.

[2] For common objects, we use the terms *quality* and *type* interchangeably.

private types and multiple outside options. This introduces a trade-off between incentives and efficient allocation for the planner who needs to decide whether to assign objects to agents with relatively high outside options. Details about the technical challenges and our proof approach are discussed in Section 3 and Subsection 4.1, respectively.

A building block in the construction is a positive assortative allocation rule, which assigns objects in a positive assortative (greedy) manner under the condition that no agent is assigned to an object of quality lower than her outside option. Then, we convert this positive assortative allocation to the optimal interim allocation rule, which may have to pool common object types (potentially together with an agent's outside option). Finally, we prove that the same mechanism remains incentive compatible (and thus, optimal) in the case of downward-private outside options.

The structure of the optimal mechanism implies that several other simple and intuitive approaches would be suboptimal. For example, one possibility is to prioritize agents based on their outside options, e.g., by disfavoring agents with higher outside options. Nevertheless, the optimal solution treats – in terms of allocation, but not in terms of payment – the highest types of agents similarly, regardless of their outside option. In contrast, the optimal solution distinguishes between lower type agents.

Some of the other notable features of the optimal solution are that (i) its participation thresholds are weakly lower than the thresholds determined by the positive assortative allocation, (ii) while in expectation an agent who participates receives a payoff at least equal her outside option, this may not hold ex post, and (iii) the cost incurred by agents of the same type but different outside options is decreasing in their outside option value.

Related literature. This paper contributes to the literature on (nonmonetary) mechanism design and costly signaling (e.g., (Gorokh et al., 2019; Toikka, 2011; Hartline and Roughgarden, 2008; Condorelli, 2012; Balseiro et al., 2019; Hoppe et al., 2009; Akbarpour et al., 2020; Chakravarty and Kaplan, 2013; Ashlagi and Shi, 2016; McAfee and McMillan, 1992; Yoon, 2011; Ashlagi et al., 2020)). Numerous studies find that pooling is sometimes required in order to reduce unnecessary effort exerted by agents; our result shows how the construction changes in the presence of outside options.

Hartline and Roughgarden (2008) characterize the Bayesian money-burning mechanism achieving the optimal social welfare in a model with multiple agents with independently and identically distributed (i.i.d.) valuations and multiple units of a single good. Condorelli (2012) generalizes this result to multiple object types. Drawing upon standard ironing techniques as in (Myerson, 1981), both show that the optimal solution might involve some type of lottery or pooling, either in terms of object qualities or agent types depending on the hazard rate of the distribution. Ashlagi et al. (2020) use results from majorization theory (Kleiner et al., 2020) to extend those results to supermodular utilities and draw connections between optimal steady-state allocation mechanisms and information design. Nevertheless, the aforementioned papers do not consider heterogeneous

outside options, and – as we discuss in Subsection 4.1 in greater detail – those techniques do not directly apply to a framework with multiple outside options.

Our work makes progress on the problem of optimal mechanism design with heterogeneous outside options in nonmonetary settings (Arnosti and Shi, 2020; Anunrojwong et al., 2020; Neilson et al., 2020; Shorrer, 2019). Arnosti and Shi (2020) consider a dynamic model of wait-lists for affordable housing when agents have horizontal preferences and type-independent outside options. They find an equivalence between two common approaches: using independent lotteries for each item, and using a wait-list in which agents lose priority when they reject an offer. They also study the trade-off between matching (i.e., assigning developments to agents with high valuation) and targeting (i.e., serving agents with the worst outside options), and find that when there are many agents with good outside options, and the distribution of outside options has a light (heavy) left tail, it is more important to match (target) well. Instead our setting assumes common qualities and we characterize the optimal mechanism.

Heterogeneous outside options have also been studied in stylized nonmonetary settings that are quite distant from ours. Anunrojwong et al. (2020) study information design in an unobservable first-come-first-served queue offering a single type of service. Each user has a private outside option of two possible types (indicating high or low need) and decides whether to wait for the service, or take her outside option. Neilson et al. (2020) compare prevalent school choice mechanisms when students may have access to different outside options. In a related admissions setting, Shorrer (2019) further characterizes the application patterns of students who differ in their outside options.

Finally, there is also a line of work (e.g., (Jehiel et al., 1996; Figueroa and Skreta, 2009; Krishna and Perry, 1997)) which considers outside options in various types of monetary auctions. However, the focus is on revenue maximization, and the outside options of individuals who are not assigned to an object are not included in the objective function.

Organization of the paper. Section 2 introduces our model and basic technical tools. Section 3 formally states our main results and describes the structure of the optimal mechanism. In Section 4 we present a proof overview, followed with the formal proof. Section 5 presents the proof for the case of downward-private outside options. Finally, Section 6 discusses practical implications of our work.

2 Model and Preliminaries

2.1 Model

Agents and common objects. We consider a continuum model of *agents* and indivisible *common objects*. The total mass of common objects equals $N < 1$ and the mass of agents is normalized to 1.[3] Every common object has a quality that belongs to a finite set $\Omega = \{\omega_1, \ldots, \omega_n\}$, where $\omega_1, \ldots, \omega_n$ are distinct nonnegative real numbers ordered increasingly. The mass of objects of quality ω is given by $F(\omega)$; in total, $N = \sum_{\omega \in \Omega} F(\omega)$.

Agents are risk-neutral and have unit demand for common objects. Every agent is characterized by a private type θ that is independently and identically distributed according to a continuous CDF G with support $\Theta = [0, 1]$. The utility that an agent of type θ derives from being assigned to an object of type ω is defined by $u(\theta, \omega) = \theta\omega$. Thus, one may interpret θ as the agent's willingness to pay for quality.

Heterogeneous outside options. Every agent also has access to an *outside option* β. Initially we assume that β is observable and later consider outside options that are downward-private (see Section 3). The agents' outside options are drawn independently from a distribution H with finite support $B = \{\beta_1, \ldots, \beta_l\}$, where β_i's are increasing in i. We interpret β as the quality of the agent's outside option. Thus, the utility that an agent of type θ and outside option β derives from being assigned to her outside option is $u(\theta, \beta)$.

For simplicity, we use the term *object* to refer to both common objects and outside options, i.e., the set $\Omega \cup B$.

Costly signals. Each agent can signal her willingness to pay for quality by exerting costly effort. Thus, if an agent of type θ is allocated a common object of quality ω at an effort cost c, then her payoff from getting assigned to a common object ω is quasi-linear in her effort cost, thus it equals $u(\theta, \omega) - c$.

Mechanisms. Appealing to the revelation principle, we focus on the class of *incentive compatible direct allocation* mechanisms without loss of generality. Such a mechanism, typically denoted by $\mathcal{M}$, elicits from every agent her type and assigns her a joint distribution over the object that she would allocate and the cost that she would incur. For an agent of type $\theta \in \Theta$ and outside option $\beta \in B$, let $X_{\mathcal{M}}(\theta, \beta)$ denote the expected object type assigned to her by the mechanism.[4] Similarly, let $c_{\mathcal{M}}(\theta, \beta)$ denote the expected cost incurred by the agent under mechanism $\mathcal{M}$; we impose $c_{\mathcal{M}}(\theta, \beta) \geq 0$ for any θ, β. Since agents are expected utility maximizers and their payoffs are

[3] For clarity of exposition, we assume that $N < 1$. We remark that, with minor changes, the proofs can be extended to the case of $N \geq 1$ as well.

[4] Note that $X_{\mathcal{M}}(\theta, \beta)$ may assign with positive probability the outside option β to the agent.

quasi-linear in costs, without loss of generality we use the tuple $(X_{\mathcal{M}}, c_{\mathcal{M}})$ to denote a mechanism $\mathcal{M}$. We call the functions $X_{\mathcal{M}}$ and $c_{\mathcal{M}}$ the *interim allocation rule* and *interim payment rule*, respectively. For simplicity, we may drop the subscript $\mathcal{M}$ when it is clearly known from the context.

The interim expected payoff of an agent of private type θ and outside option β who reports θ' is

$$\Phi^{\beta}_{\mathcal{M}}(\theta;\theta') \triangleq u(\theta, X_{\mathcal{M}}(\theta',\beta)) - c_{\mathcal{M}}(\theta',\beta) = \theta X_{\mathcal{M}}(\theta',\beta) - c_{\mathcal{M}}(\theta',\beta).$$

A mechanism is *incentive-compatible* (IC) if the expected payoff of every participating agent is maximized when she reports her true type to the mechanism. A mechanism is *individually rational* (IR) if the expected payoff of every participating agent is at least equal to her payoff from being assigned to her outside option at no cost. By the revelation principle, we require that any mechanism $\mathcal{M}$ must satisfy the interim *incentive compatibility* and *individual rationality* constraints, i.e.,

$$\Phi^{\beta}_{\mathcal{M}}(\theta;\theta) \geq \Phi^{\beta}_{\mathcal{M}}(\theta;\theta'), \forall \theta, \theta' \in \Theta, \forall \beta \in B, \tag{2.1}$$

$$\Phi^{\beta}_{\mathcal{M}}(\theta;\theta) \geq \theta\beta, \forall \theta \in \Theta, \forall \beta \in B. \tag{2.2}$$

We assume that an agent who is indifferent between taking her outside option and what the mechanism assigns her, always chooses the former.[5]

Social welfare. The planner's objective is to choose an incentive-compatible and individually rational mechanism with the highest *social welfare*, where the *social welfare* in a mechanism $\mathcal{M}$ is defined by $W_{\mathcal{M}} = \sum_{\beta \in B} H(\beta) \mathbb{E}_{\theta \sim G}\left[\Phi^{\beta}_{\mathcal{M}}(\theta;\theta)\right]$. An individually rational and incentive-compatible mechanism $\mathcal{M}^*$ is *optimal* if there is no other mechanism $\mathcal{M}$ such that $W_{\mathcal{M}^*} < W_{\mathcal{M}}$.

2.2 Technical preliminaries

The following technical definitions can be skipped by readers familiar with them.

Ironing. Adopting the formulation of Myerson (1981), let $\psi : \Theta \to \mathbb{R}$ denote the *virtual payoff function*

$$\psi(\theta) = \frac{1 - G(\theta)}{g(\theta)}. \tag{2.3}$$

This quantity is also known as the *inverse hazard rate* function.

Definition 1 (Ironed virtual payoff function (Hartline and Roughgarden, 2008; Condorelli, 2012))**.** *The monotone nondecreasing* ironed virtual payoff function $\bar{\psi}$ *is defined as follows. For each*

[5] In the optimal mechanism, a measure zero of the agents would be indifferent.

$q \in [0,1]$, *let*

$$K(q) = \int_0^q \frac{1-z}{g(G^{-1}(z))}\,\mathrm{d}z.$$

Define $J(q)$, $q \in [0,1]$ *as the* convex hull *of* $K(q)$.[6] *For all* $\theta \in \Theta$, *let*

$$\overline{\psi}(\theta) = J'(G(\theta)).^{7}$$

We say that any maximal interval $I = [\theta_1, \theta_2] \subseteq \Theta$ such that $J(G(\theta)) < K(G(\theta))$, for all $\theta \in (\theta_1, \theta_2)$ is an *ironed interval* of ψ.

Fan-Lorentz inequality. For two nondecreasing functions $Y, Z \in L^1(0,1)$, we say that Y *majorizes* Z (and write $Y \prec Z$) if

$$\int_q^1 Y(s)\,\mathrm{d}s \le \int_q^1 Z(s)\,\mathrm{d}s, \forall q \in [0,1],$$
$$\int_0^1 Y(s)\,\mathrm{d}s = \int_0^1 Z(s)\,\mathrm{d}s.$$

The following property will be useful. We use the notation $\underline{\omega} = \min\{\beta_1, \omega_1\}$, $\overline{\omega} = \max\{\beta_l, \omega_n\}$.[8]

Theorem 1 (Fan and Lorentz (1954))**.** *Let* $V : [\underline{\omega}, \overline{\omega}] \times [0,1] \to \mathbb{R}$ *and let* $Y, Z : [0,1] \to [\underline{\omega}, \overline{\omega}]$ *be such that* $Y \prec Z$. *Then*

$$\int_0^1 V(Y(q), q)\,\mathrm{d}q \le \int_0^1 V(Z(q), q)\,\mathrm{d}q$$

holds for any nondecreasing functions $Y, Z : [0,1] \to [\underline{\omega}, \overline{\omega}]$ *such that* $Y \prec Z$ *if and only if the function* $V(x,q)$ *is convex in* x *and supermodular in* (x,q).

Auxiliary definitions for CDFs. Given a continuous CDF $G : [0,1] \to [0,1]$, let $\overline{G}(\theta) = 1 - G(\theta)$ denote its corresponding survival function. Let also $\overline{G}^{-1}(q)$, $q \in [0,1]$, denote the inverse of $\overline{G}$.

Finally, for the discrete function $F : \Omega \to [0,N]$, we define

$$\overline{F}^{-1}(q) = \inf\left\{\omega \in \Omega : \sum_{\omega' \in \Omega:\ \omega \ge \omega'} F(\omega') \ge N - q\right\}, q \in [0,N].$$

[6] That is, $J(q) = convK(q) = \min\{yK(q_1) + (1-y)K(q_2) \mid y \in [0,1], q_1, q_2 \in [0,1], yq_1 + (1-y)q_2 = q\}$.

[7] If $J'(q)$ does not exist at some $q \in [0,1]$, extend the definition of $\overline{\psi}$ by right-continuity.

[8] We state the theorem as in Kleiner et al. (2020).

3 Optimal mechanism design: Main results

In this section we present our two main results and discuss the technical challenges of the problem. Theorem 2 characterizes the optimal mechanism $\mathcal{M}^*$ under the assumption that outside options are public (i.e., known to the planner). Theorem 3 shows that $\mathcal{M}^*$ is also optimal in the case of downward-private options.

Public outside options. Towards characterizing $\mathcal{M}^*$ we first construct an auxiliary *positive assortative* interim allocation rule, namely X_{P}, which we next describe intuitively and then formally. The allocation rule X_{P} assigns common objects in a greedy manner which assures that higher-type agents are given access to common objects with higher qualities, irrespective of their outside option. Following a decreasing order of object qualities, X_{P} assigns the next available common object to agents in decreasing order of their type, while ensuring that no agent receives an object with a quality lower than her outside option. Hence, for every outside option β_i, X_{P} determines a *participation threshold* γ_i^{P}, which corresponds to the minimum agent type with outside option β_i who allocates a common object with quality higher than β_i in this greedy process.

To formalize this, define first by ν_i the index of the lowest quality object for an agent with outside option β_i, i.e.,

$$\nu_i = \min\{j \leq n+1 : \omega_j > \beta_i\}$$

and for convenience define $\nu_{l+1} = n+1$. For each $1 \leq i \leq l$, X_{P} is formally defined as

$$X_{\mathrm{P}}(\theta, \beta_i) = \begin{cases} \hat{X}_{\mathrm{P}}(\theta), & \theta > \gamma_i^{\mathrm{P}}, \\ \beta_i, & \theta \leq \gamma_i^{\mathrm{P}}, \end{cases} \tag{3.1}$$

where

$$\hat{X}_{\mathrm{P}}(\theta) = \overline{F}^{-1}\left(\sum_{k=i+1}^{l}\sum_{j=1}^{k} H(\beta_j)\left(\overline{G}(\gamma_k^{\mathrm{P}}) - \overline{G}(\gamma_{k+1}^{\mathrm{P}})\right) + \sum_{j=1}^{i} H(\beta_j)\left(\overline{G}(\theta) - \overline{G}(\gamma_{i+1}^{\mathrm{P}})\right)\right), \gamma_i^{\mathrm{P}} < \theta \leq \gamma_{i+1}^{\mathrm{P}},$$

$$\gamma_{l+1}^{\mathrm{P}} = 1,$$

$$\gamma_i^{\mathrm{P}} = \overline{G}^{-1}\left(\min\left\{1, \frac{\sum_{\nu_i \leq j < \nu_{i+1}} F(\omega_j)}{\sum_{j=1}^{i} H(\beta_j)} + \overline{G}(\gamma_{i+1}^{\mathrm{P}})\right\}\right), 1 \leq i \leq l.$$

To simplify the definition of the optimal mechanism, we introduced the notation $\omega_{n+1} = \beta_l + \Delta$ for an arbitrary $\Delta > \omega_n$ and $F(\omega_{n+1}) = 0$.[9]

Given X_{P}, we next intuitively describe how to construct the optimal interim allocation rule $X_{\mathcal{M}^*}$ (see (3.2)), and how to show its optimality. We first find the ironed intervals of ψ through the

[9] The artificial common object quality w_{n+1} of zero capacity helps define the threshold $\gamma_i^{\mathrm{P}} = 1$ for the case $\beta_i > \omega_n$, $i \leq l$.

ironing process in Definition 1 which transforms any virtual payoff function $\psi(\theta) = \frac{1-G(\theta)}{g(\theta)}$ that may not be nondecreasing to a nondecreasing ironed virtual payoff function $\overline{\psi}$. If ψ is nondecreasing everywhere, then $X_{\mathcal{M}^*}$ coincides with X_{P}. However, this is not the case in general; thus, there exists a collection of ironed intervals of ψ, namely $(\underline{\theta}_c, \bar{\theta}_c)$ indexed by $c \in \mathcal{C}$. To find the optimal mechanism $\mathcal{M}^*$ we need to carefully pool objects across each such interval in a way that the pooling respects the agents' individual rationality.

The allocation of common object types is the same for agents with the same type θ (and possibly different outside options β) except for two notable cases. The first one is that agents with outside option β_i and types $\theta \leq \gamma_i^{\mathrm{P}}$ are not assigned to any common object and get their outside option. The second case is when $\gamma_i^{\mathrm{P}} \in (\underline{\theta}_c, \bar{\theta}_c)$ for some $c \in \mathcal{C}$. In this case $X_{\mathrm{P}}(\cdot, \beta_i)$ would assign common objects to agent types $\theta > \gamma_i^{\mathrm{P}}$ and their outside option β_i to $\theta \leq \gamma_i^{\mathrm{P}}$. However, ironing together with incentive compatibility require $\mathcal{M}^*$ to treat all agents with the same outside option and with types belonging to $(\underline{\theta}_c, \bar{\theta}_c)$ identically. Thus, $\mathcal{M}^*$ would pool the objects (including the outside option β_i) assigned by X_{P} to the agents who have a type belonging to that interval and an outside option β_i. This property would imply that the participation thresholds of $X_{\mathcal{M}^*}$ are weakly lower than those of X_{P}, i.e., $\gamma_i^{\mathcal{M}^*} \leq \gamma_i^{\mathrm{P}}$.

We next characterize the optimal mechanism for the case where outside options are known to the planner. Theorem 2 is our first main result.

Theorem 2. *Consider the collection of maximal ironed intervals of ψ, namely $(\underline{\theta}_c, \bar{\theta}_c)$ indexed by $c \in \mathcal{C}$. For each $\beta_i \in B$, let*

$$X_{\mathcal{M}^*}(\theta, \beta_i) = \begin{cases} X_{\mathrm{P}}(\theta, \beta_i), & \theta \notin \cup_{c \in \mathcal{C}} (\underline{\theta}_c, \bar{\theta}_c] \\ \frac{\int_{\underline{\theta}_c}^{\bar{\theta}_c} X_{\mathrm{P}}(z, \beta_i)\, \mathrm{d}G(z)}{G(\bar{\theta}_c) - G(\underline{\theta}_c)}, & \theta \in (\underline{\theta}_c, \bar{\theta}_c], \end{cases} \tag{3.2}$$

$$c_{\mathcal{M}^*}(\theta, \beta_i) = \theta X_{\mathcal{M}^*}(\theta, \beta_i) - \int_0^{\theta} X_{\mathcal{M}^*}(z, \beta_i)\, \mathrm{d}z. \tag{3.3}$$

Then, the optimal mechanism is given by $(X_{\mathcal{M}^*}, c_{\mathcal{M}^*})$.

We remark that our proof also provides an algorithm to construct the optimal mechanism by providing computable expressions (3.2) and (3.3) for the allocation and payment rules, respectively. We next use an example to demonstrate the optimal mechanism.

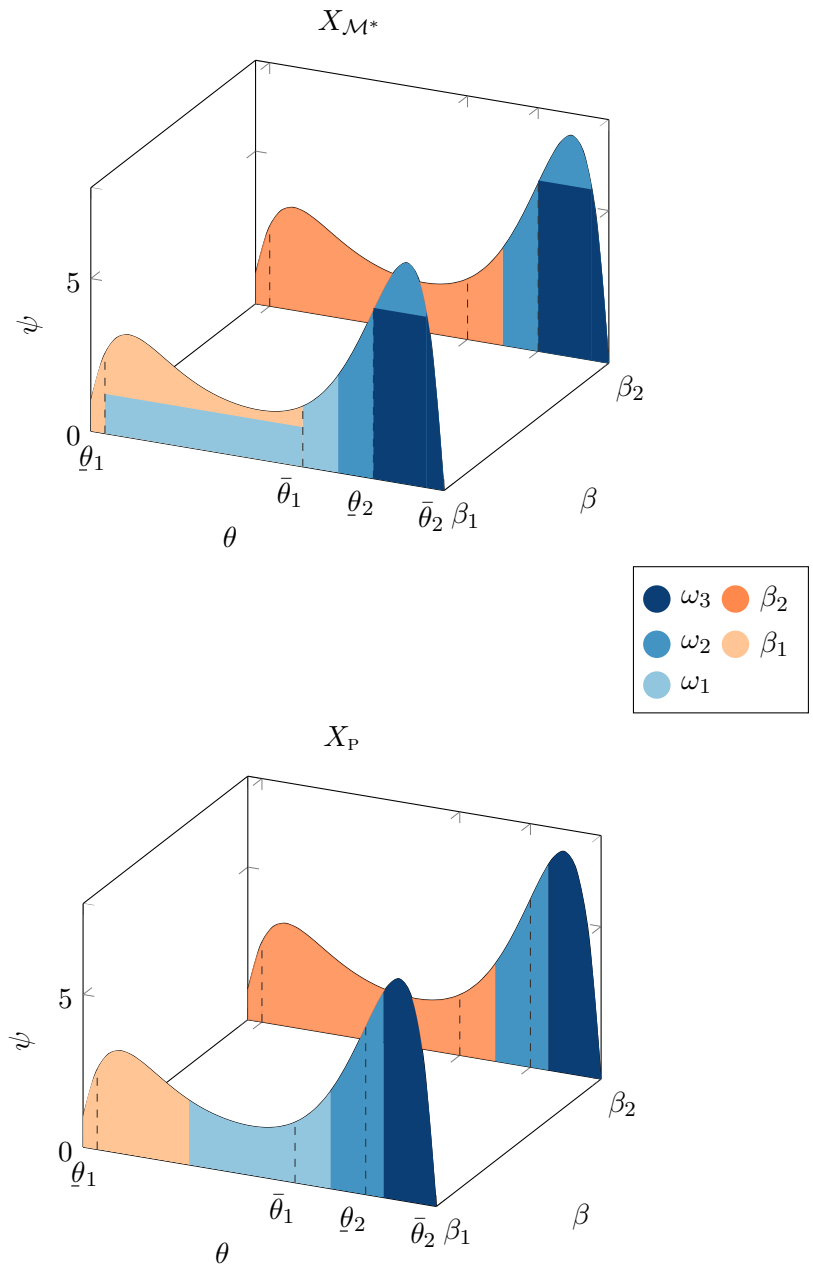


Figure 1: Optimal interim allocation rule $X_{\mathcal{M}^*}(\theta, \beta)$ (**top**) and ironed-optimal interim allocation rule $X_{\mathrm{P}}(\theta, \beta)$ (**bottom**) for Example 1.

Example 1. *Figure 1 illustrates the structure of the optimal mechanism through a hypothetical example with 2 outside options and 3 object qualities such that* $\beta_1 < \omega_1 < \beta_2 < \omega_2 < \omega_3$. *As detailed next, the top plot depicts* ψ *and the optimal allocation* $X_{\mathcal{M}^*}(\theta, \beta)$, *which differs between agents with outside options* β_1 *and* β_2. *The bottom plot depicts* $X_{\mathrm{P}}(\theta, \beta)$.

The virtual payoff function ψ *has two ironed intervals:* $(\underline{\theta}_1, \bar{\theta}_1)$ *and* $(\underline{\theta}_2, \bar{\theta}_2)$. *In each of those*

intervals, $X_{\mathcal{M}^*}(\cdot, \beta_1)$ and $X_{\mathcal{M}^*}(\cdot, \beta_2)$ are constant. For the right ironed interval $(\underline{\theta}_2, \bar{\theta}_2)$, all objects of the highest quality ω_3 and some objects of the second-highest quality ω_2 are assigned uniformly at random to agents with types $\theta \in (\underline{\theta}_2, \bar{\theta}_2]$; observe that both groups of agents (β_1, β_2) get the same expected object type in that case. For the left ironed interval $(\underline{\theta}_1, \bar{\theta}_1)$, agents with outside option β_2 get their outside option, since $\beta_2 > \omega_1$. In contrast, agents with type belonging to $(\underline{\theta}_1, \bar{\theta}_1)$ but with outside option β_1 are assigned uniformly at random to the remaining common objects of quality ω_1 and their outside option $\beta_1 < \omega_1$, both with some positive probability. Thus, agents with β_1 have a lower participation threshold $\gamma_1^{\mathcal{M}^*} = \underline{\theta}_1$ compared to agents with β_2 and $\bar{\theta}_1 < \gamma_2^{\mathcal{M}^*} < \underline{\theta}_2$.

Discussion of technical challenges and inapplicability of existing frameworks. The main technical challenge stems from having *both* heterogeneous object types (with fixed capacities) and heterogeneous outside options, which implies that the rule $X(\theta, \beta)$ depends on *both* θ and β. Variants of the problem have been studied in the case of a single outside option typically normalized to 0 (see, e.g., (Condorelli, 2012; Akbarpour et al., 2020; Ashlagi et al., 2020) for multiple object types, and (Hartline and Roughgarden, 2008; Toikka, 2011) for a single object type). In a dynamic matching model with a single outside option, Ashlagi et al. (2020) build upon the extreme-point characterization results of (Kleiner et al., 2020) to prove the optimality of mechanisms that may pool objects. Their results, however, do not apply to our setting due to the multidimensionality of the allocation rule here. This is because the extreme point characterization results of (Kleiner et al., 2020) are based on the single-dimensional majorization property.

Downward-private outside options. Our framework finds a natural application to public allocation systems without money, such as public or affordable housing. For instance, in New York people with low or moderate income can obtain affordable housing through the New York City Housing Authority (NYCHA).[10] In such systems, a social planner needs to successfully distinguish individuals with high and low need in terms of their available outside option. For example, two individuals may value the same types of housing similarly (same type θ) but one of them may be homeless while the other might have the option to be hosted by another family member.

If the outside options of the participants are not observed, one valid concern of the social planner may be that certain agents might misreport their outside option. Authorities administrating such public allocation systems, however, may ask for supporting evidence about each individual's reported outside option. Then, while individuals may be able to *deflate* such evidence (e.g., by withholding information about their available options), they may not be able inflate it (i.e., providing evidence for out-of-reach options). Returning to our housing example, consider two categories of outside options among individuals who cannot afford their current rent: having the option of staying at their family house (β_2), or not (β_1). Under our assumption, an applicant who does have this option may pretend not to have it, but not vice versa.

[10] See https://access.nyc.gov/programs/public-housing/ for details.

Here, instead of assuming that outside options are known to the planner, we assume that they are *downward-private*, formally meaning that an agent with true outside option β can report any $\beta' \leq \beta$ to the mechanism but not $\beta' > \beta$. We will show that in the generalized setting with downward-private outside options, our mechanism $\mathcal{M}^*$ from Theorem 2 remains optimal.

Recall that both the allocation and payment rules already depended on θ, β in the previous section. In the presence of downward-private types, they need to satisfy stronger incentive compatibility constraints. Formally, let $\Phi_{\mathcal{M}^*}(\theta, \beta; \theta', \beta')$ denote the interim expected payoff of an agent of private type θ and outside option β who reports θ' and $\beta' \leq \beta$. The agent's payoff is computed assuming that if she misreports (her outside option) and is assigned to her outside option by the mechanism, then she attains her *true* outside option, and thus utility $u(\theta, \beta)$. The incentive compatibility and individual rationality constraints respectively are

$$\Phi_{\mathcal{M}^*}(\theta, \beta; \theta, \beta) \geq \Phi_{\mathcal{M}^*}(\theta, \beta; \theta', \beta'), \forall \theta, \theta' \in \Theta, \forall \beta' < \beta \in B, \tag{3.4}$$

$$\Phi_{\mathcal{M}^*}(\theta, \beta; \theta, \beta) \geq \theta\beta, \forall \theta \in \Theta, \forall \beta \in B. \tag{3.5}$$

Our second main result is the following.

Theorem 3. *When the outside options are downward-private, the mechanism $\mathcal{M}^*$, given by (3.2)-(3.3), is incentive-compatible, individually rational, and optimal.*

We build upon the proof of Theorem 2 to prove Theorem 3. We show that for any fixed $\theta \in \Theta$, the payment rule $c_{\mathcal{M}^*}(\theta, \beta)$ is nonincreasing in $\beta \in B$. Furthermore, we show that an agent who reports an outside option β' lower than her true one β attains the same utility under both reports (holding her type report fixed). These two ideas allow us to show that $\mathcal{M}^*$ remains incentive-compatible and individually rational in the setting with downward-private outside options. Therefore, $\mathcal{M}^*$ remains optimal. The details are discussed in Section 5.

4 Public Outside Options: Proof of Theorem 2

We next describe the key steps of the proof of Theorem 2 here; the full proof is presented sequentially in Subsections 4.2-4.4. We note that a reader familiar with the literature may wish to skip Subsection 4.2, since it is based on standard techniques, and proceed to Subsection 4.3.

4.1 Proof overview

The proof has two main steps. In the first step, we consider an upper bound to our problem, namely maximizing the ironed virtual payoff. We show that the solution to this problem is a *partially symmetric* allocation rule, i.e., it is a monotone rule that allocates the same expected object type $\hat{X}(\theta)$ to all agents θ with outside option $\beta < \hat{X}(\theta)$ but assigns the rest of the agents

to their outside option. Moreover, we show that the positive assortative allocation rule X_{P} is the solution to this maximization problem. In the second step, we convert X_{P} to the optimal allocation rule as defined in Theorem 2. The first and second step are presented in Subsections 4.3 and 4.4, respectively, and are outlined below.

The first step involves several technical lemmas. We first show that the allocation rule that maximizes ironed virtual payoff does not feature pooling (unlike the optimal rule $X_{\mathcal{M}^*}$ which may need to pool). This result follows from an application of the Fan-Lorentz inequality (Theorem 1) on an appropriately defined family of allocation rules given by a majorization constraint (Lemma 4).

We proceed to show that agents with higher outside options have higher participation thresholds, i.e., $\gamma_1 \leq \gamma_2 \leq \ldots \leq \gamma_l$ (see Lemma 5). We prove this through a *transport* argument that transforms a given allocation rule X monotone in θ to an intermediate rule $\tilde{\tilde{X}}$ which indeed has increasing participation thresholds but may not be monotone in θ (and thus not incentive-compatible). We then transform $\tilde{\tilde{X}}$ to a nondecreasing interim allocation rule $\tilde{X}$ by applying an ironing procedure—this time on the allocation rule itself. The result is a new rule $\tilde{X}$ which is monotone in θ and has increasing participation thresholds.

Recall that in the first step we aim to show that the optimal allocation rule is partially symmetric. We achieve this by transforming an allocation rule that maximizes ironed virtual payoff and has increasing participation thresholds to a partially symmetric allocation rule (see Lemma 6). This is done by a local averaging argument over each interval $(\gamma_{i-1}, \gamma_i]$; the result would be an allocation rule which satisfies all properties for being partially symmetric but monotonicity. We use another ironing procedure on this rule to transform it to a monotone rule.

In the final part of the argument in the first step, we maximize ironed social welfare over the class of partially symmetric rules. The optimal rule turns out to be the positive assortative rule X_{P}.

In the second step (Subsection 4.4), we transform X_{P} to an optimal allocation rule $X_{\mathcal{M}^*}$. This is done by pooling object types in ironed intervals. As maximizing ironed virtual payoff gives an upper bound for optimal social welfare, and since $X_{\mathcal{M}^*}$ attains this upper bound, we conclude that it is the optimal interim allocation rule for our social welfare objective.

4.2 Characterizing the optimal mechanism à la Myerson

In this section, we apply classic arguments (Myerson, 1981) to show that the optimal allocation rule must be monotone in $\theta \in \Theta$ for each fixed outside option $\beta \in B$, characterize the payment rule, and reduce our social welfare maximization problem to an (ironed) virtual payoff optimization problem.

First, we start with a simple observation. When an incentive compatible and individually rational mechanism $\mathcal{M}$ assigns $X_{\mathcal{M}}(\theta, \beta) \leq \beta$ to some agent with type θ and outside option β, we can replace $\mathcal{M}$ with another incentive compatible and individually rational mechanism $\mathcal{M}'$ such

that $X_{\mathcal{M}'}(\theta,\beta) = \beta$ and $c_{\mathcal{M}'}(\theta,\beta) = 0$. In both $\mathcal{M}$ and $\mathcal{M}'$, those agents choose their outside option β and contribute $\Phi^{\beta}_{\mathcal{M}}(\theta;\theta) = \Phi^{\beta}_{\mathcal{M}'}(\theta;\theta) = \theta\beta$ to the social welfare. This follows immediately from the individual rationality constraint in (2.2) and the nonnegativity of payments. Thus, we summarize this observation in the following lemma:

Lemma 1. *For any mechanism $\mathcal{M}$, there exists a mechanism $\mathcal{M}'$ such that*

$$X_{\mathcal{M}'}(\theta,\beta) = \max\{X_{\mathcal{M}}(\theta,\beta),\beta\}$$
$$c_{\mathcal{M}'}(\theta,\beta) = \begin{cases} c_{\mathcal{M}}(\theta,\beta), & \text{if } X_{\mathcal{M}}(\theta,\beta) > \beta \\ 0, & \text{if } X_{\mathcal{M}}(\theta,\beta) \leq \beta, \end{cases}$$

which yields equal social welfare to $\mathcal{M}$.

The following lemma is a standard adaption of Myerson's lemma to our setting; it characterizes the set of incentive-compatible and individually rational mechanisms and establishes the existence of participation thresholds for agents of different outside options. The proof uses common arguments mirroring Myerson (1981) and can be found in Appendix A together with other omitted proofs.

Lemma 2. *Let $\mathcal{M}$ be an incentive-compatible and individually rational mechanism $\mathcal{M}$. Then, $X_{\mathcal{M}}(\theta,\beta_i)$ and $c_{\mathcal{M}}(\theta,\beta_i)$ satisfy the following properties:*

(i) for each fixed $\beta_i \in B$, $X_{\mathcal{M}}(\theta,\beta_i)$ is nondecreasing in $\theta \in \Theta$; and

(ii) for all $\theta \in \Theta$, $\beta_i \in B$,

$$c_{\mathcal{M}}(\theta,\beta_i) = \theta X_{\mathcal{M}}(\theta,\beta_i) - \int_0^{\theta} X_{\mathcal{M}}(z,\beta_i)\,\mathrm{d}z. \tag{4.1}$$

Furthermore, for each $1 \leq i \leq l$, there exists a participation threshold $\gamma_i^{\mathcal{M}}$ such that $X_{\mathcal{M}}(\theta,\beta_i) > \beta_i$ for all $\theta > \gamma_i^{\mathcal{M}}$ and $X_{\mathcal{M}}(\theta,\beta_i) = \beta_i$ for all $\theta \leq \gamma_i^{\mathcal{M}}$.

We can now characterize the set of feasible interim allocation rules (and thus their corresponding interim payment rule using (4.1)). First, let $\mathcal{D}$ denote the set of all feasible probability distributions $x(\theta,\beta,\cdot) \in \Delta(\Omega \cup \{\beta\})$, defined for each $\theta \in \Theta$, $\beta \in B$, that respect the available mass $F(\omega)$ of each common object type $\omega \in \Omega$, i.e.,

$$\mathcal{D} = \{x : \Theta \times B \times (\Omega \cup B) \to [0,1] \mid x(\theta,\beta,\cdot) \in \Delta(\Omega \cup \{\beta\}), \sum_{\beta \in B} H(\beta) \int_0^1 x(\theta,\beta,\omega)\,\mathrm{d}G(\theta) \leq F(\omega), \forall \omega \in \Omega\}. \tag{4.2}$$

Then, the set $\mathcal{S}$ of all *feasible interim allocation rules* $X(\theta,\beta)$, $\theta \in \Theta$, $\beta \in B$, in a setting with

heterogeneous outside options is given by

$$\mathcal{S} = \{X : \Theta \times B \to [\omega_1, \omega_n] \mid X(\theta, \beta) \geq \beta \text{ is nondecreasing in } \theta \text{ and } \exists x \in \mathcal{D} \text{ s.t. } X(\theta, \beta) = \sum_{\omega \in \Omega \cup \{\beta\}} \omega x(\theta, \beta, \omega)\}. \tag{4.3}$$

Given any feasible rule $X \in \mathcal{S}$, Lemma 2 together with an application of Fubini's theorem implies that the social welfare of a mechanism $\mathcal{M}$ can be written in terms of the virtual payoff function $\psi(\theta)$, i.e.,

$$\sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) X_{\mathcal{M}}(\theta, \beta_i) \, \mathrm{d}G(\theta).$$

Thus, the social welfare under any feasible interim allocation rule $X \in \mathcal{S}$ can be expressed in terms of the virtual payoff $\psi(\theta)$ and our optimization problem can be written as follows.

Proposition 1. *The social welfare $W_{\mathcal{M}^*}$ of the optimal mechanism $\mathcal{M}^*$ satisfies*

$$W_{\mathcal{M}^*} = \sup_{X \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta). \tag{4.4}$$

If the virtual payoff function $\psi(\theta)$ is nondecreasing, then later in Lemma 7 we will show that a positive assortative assignment of object to agents (i.e., rule X_{P}) is optimal. For the general case where $\psi(\theta)$ is not nondecreasing, we adopt the *ironing* process of Myerson (1981) (see, e.g., Hartline and Roughgarden (2008); Toikka (2011); Condorelli (2012), for analogous techniques) in order to transform $\psi(\theta)$ to the *ironed virtual payoff function* $\overline{\psi}(\theta)$, which is nondecreasing in $\theta \in \Theta$. The reader may refer to Definition 1 for the exact construction of $\overline{\psi}$.

Note that, since J is the convex hull of K, $J(G(\theta))$ and $K(G(\theta))$ coincide at the endpoints of any ironed interval. Moreover, by the convexity of J, it follows that the function $\overline{\psi}(\theta)$ is nondecreasing. Furthermore, it holds that

$$\overline{\psi}(\theta) = \psi(\theta) + J'(G(\theta)) - K'(G(\theta)). \tag{4.5}$$

The next lemma generalizes the result for separable objectives in Toikka (2011, Theorem 3.8) to our setting with multiple outside options and multiple common object types with fixed capacities. For completeness, we provide the proof in Appendix A.

Lemma 3. *For any $X \in \mathcal{S}$ and $\beta_i \in B$, it holds that*

$$\int_0^1 \psi(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta) \leq \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta). \tag{4.6}$$

If $X(\theta, \beta_i)$ is constant in each ironed interval of ψ, then

$$\int_0^1 \psi(\theta, \beta_i) X(\theta, \beta_i) \, \mathrm{d}G(\theta) = \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta). \tag{4.7}$$

Conversely, if for some $X \in \mathcal{S}$ it holds that

$$\sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta) = \max_{Y \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) Y(\theta, \beta_i) \, \mathrm{d}G(\theta), \tag{4.8}$$

then for all $\beta_i \in B$, $X(\theta, \beta_i)$ is constant in any ironed interval of ψ.

Eq. (4.6) implies that the optimal expected ironed virtual payoff serves as an upper bound to the optimal social welfare. Furthermore, (4.7) and (4.8) together indicate that this bound is tight. This is a key insight that we will leverage in the proof of our main result. However, after this point we can not directly solve for the optimal two-dimensional $X_{\mathcal{M}^*}$, so we need to apply the next series of results.

4.3 Ironed-optimal interim allocation rules

Instead of the actual social welfare objective, we initially focus on the *ironed social welfare* optimization problem

$$\max_{X \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta), \tag{4.9}$$

which – by Lemma 3 – gives an upper bound on the optimal social welfare $W_{\mathcal{M}^*}$ in (4.4).

We solve (4.9) and construct its optimal solution, which we call an *ironed-optimal* interim allocation rule. It turns out that the interim allocation rule of the optimal mechanism belongs to the following class $\mathcal{U}$ of *partially symmetric* rules, which allocate the same expected object type $\hat{X}(\theta)$ to all agents θ with outside option $\hat{X}(\theta) > \beta$ but assign the remaining agents to their outside option. We formally define class $\mathcal{U}$ below.

Definition 2 (Partially symmetric interim allocation rules)**.** *Let $\mathcal{U} \subseteq \mathcal{S}$ denote the class of all feasible interim allocation rules X of the form*

$$X(\theta, \beta_i) = \begin{cases} \hat{X}(\theta), & \theta > \gamma_i \\ \beta_i, & \theta \leq \gamma_i, \end{cases} \tag{4.10}$$

where γ_i, $1 \leq i \leq l$, are its corresponding participation thresholds. We call any $X \in \mathcal{U}$ a partially symmetric *interim allocation rule.*

The general idea is that, given any ironed-optimal rule $X \in \mathcal{S}$, we can construct a new rule $X^* \in \mathcal{U}$ that yields higher than or equal social welfare as X. Our next steps are as follows. In

Lemma 4, we start with a technical result showing, via an application of the Fan-Lorentz inequality, that an ironed-optimal rule $X(\theta, \beta_i)$ does *not* pool the outside option β_i and objects $\omega \in \Omega$.[11] Next, in Lemma 5 we show that, given an ironed-optimal rule $X \in \mathcal{S}$, agents with higher outside options β_i must have weakly higher participation thresholds, i.e., $\gamma_1 \leq \gamma_2 \leq \ldots \leq \gamma_l$. Using the result from Lemma 5, in Lemma 6 we apply a local averaging process in each interval $(\gamma_{i-1}, \gamma_i]$ to show that there exists a partially symmetric interim allocation rule which is ironed-optimal. Finally, we construct the interim allocation rule $X_{\mathrm{P}} \in \mathcal{U}$. In Lemma 7 and building upon the previous steps, we show that X_{P} is ironed-optimal.

Lemma 4. *Assume that $X \in \mathcal{S}$ is ironed-optimal. Then, there exists a rule $\tilde{X} \in \mathcal{S}$ with participation thresholds $\tilde{\gamma}_i$, $1 \leq i \leq l$, such that $\tilde{X}(\theta, \beta_i) \geq \min_j\{\omega_j : \omega_j > \beta_i\}$ for every $\beta_i \in B$ and $\theta > \tilde{\gamma}_i$, which yields weakly higher ironed social welfare than X.*

Proof. Fix any $\beta_i \in B$; the same process can be repeated for any other outside option. Consider the set $\mathcal{L}$ of allocation rules $\tilde{X} \in \mathcal{S}$ that allocate (a subset of) the same common objects to agents with outside option β_i as $X(\theta, \beta_i)$ (but in potentially different order); for any other $\beta \neq \beta_i$, $\tilde{X}(\theta, \beta)$ remains identical to $X(\theta, \beta)$. Formally, by the definition of $\mathcal{S}$ in (4.3), it follows that for any such $\tilde{X} \in \mathcal{L} \subseteq \mathcal{S}$, there must exist a $\tilde{x} \in \mathcal{D}$ such that $\tilde{X}(\theta, \beta_i) = \sum_{\omega \in \Omega \cup \{\beta_i\}} \omega \tilde{x}(\theta, \beta_i, \omega)$ and for all $\omega \in \Omega$,

$$\int_0^1 \tilde{x}(\theta, \beta_i, \omega)\, \mathrm{d}G(\theta) \leq \int_0^1 x(\theta, \beta_i, \omega)\, \mathrm{d}G(\theta).$$

One of those mechanisms is $\tilde{X}_{\mathcal{L},\mathrm{P}}(\theta, \beta_i)$ that greedily reallocates common object types $\omega > \beta_i$ to agent types in a positive assortative manner, while it assigns β_i to the lowest remaining agents. Formally, for all $\omega > \beta_i$, let

$$F_{\mathcal{L}}(\omega) = H(\beta_i) \int_0^1 x(\theta, \beta_i, \omega)\, \mathrm{d}G(\theta).$$

Then, $\tilde{X}_{\mathcal{L},\mathrm{P}}(\theta, \beta_i)$ is defined as

$$\tilde{X}_{\mathcal{L},\mathrm{P}}(\theta, \beta_i) = \begin{cases} \overline{F}_{\mathcal{L}}^{-1}\left(\overline{G}(\theta) H(\beta_i)\right), & \theta > \tilde{\gamma}_i \\ \beta_i, & \theta \leq \tilde{\gamma}_i, \end{cases}$$

where $\tilde{\gamma}_i \triangleq \overline{G}^{-1}\left(\frac{\sum_{\omega > \beta_i} F_{\mathcal{L}}(\omega)}{H(\beta_i)}\right)$.

Finally, we apply Theorem 1 for $V(Y(q), q) = \overline{\psi}(G^{-1}(q))Y(q)$ where $Y(q) = X(G^{-1}(q), \beta_i)$ and $\tilde{Y}_{\mathcal{L},\mathrm{P}}(q) = \tilde{X}_{\mathcal{L},\mathrm{P}}(G^{-1}(q), \beta_i)$, $q \in [0, 1]$. The function V is supermodular and convex in the first argument. Furthermore, for any feasible Y, it holds that $Y \prec \tilde{Y}_{\mathcal{L},\mathrm{P}}$. Consequently, Theorem 1 gives

[11] However, as will show in Theorem 2, the optimal rule achieving the maximum social welfare *may* indeed pool the outside option β_i of an agent and objects of higher type than β_i.

us

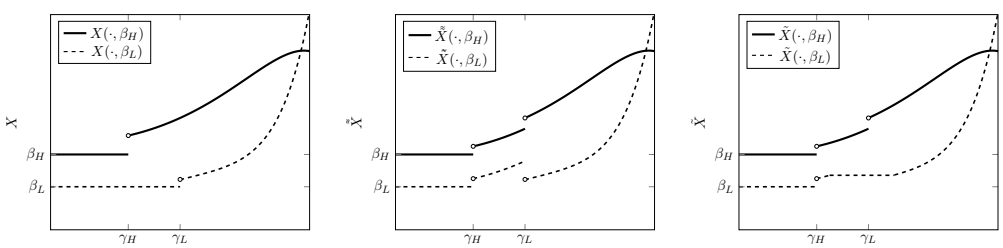
$$
\begin{aligned}
H(\beta_i)\int_0^1 \overline{\psi}(\theta)X(\theta,\beta_i)\,\mathrm{d}G(\theta) &= H(\beta_i)\int_0^1 \overline{\psi}(G^{-1}(q))Y(q)\,\mathrm{d}q \\
&\leq H(\beta_i)\int_0^1 \overline{\psi}(G^{-1}(q))\tilde{Y}_{\mathcal{L},\mathrm{P}}(q)\,\mathrm{d}q \\
&= H(\beta_i)\int_0^1 \overline{\psi}(\theta)\tilde{X}_{\mathcal{L},\mathrm{P}}(\theta,\beta_i)\,\mathrm{d}G(\theta).
\end{aligned}
$$

Thus, $\tilde{X}_{\mathcal{L},\mathrm{P}}$ achieves weakly higher ironed social welfare than X.

□

One of the key steps in our proof is Lemma 5 below, which shows that the participation thresholds are increasing in order, i.e., $\gamma_1 \leq \gamma_2 \leq \ldots \gamma_l$.

Lemma 5. *Let $X \in \mathcal{S}$ be ironed-optimal. Then, there exists a $\tilde{X} \in \mathcal{S}$ with nondecreasing participation thresholds $\gamma_1 \leq \gamma_2 \leq \ldots \leq \gamma_l$, which yields weakly higher ironed social welfare than X.*



(a) (b) (c)

Figure 2: The figure illustrates the transport and ironing arguments used in the proof of Lemma 5.

The main idea of the proof is illustrated through an example in Figure 2 (the proof is given in Appendix A). For the sake of contradiction, we assume that for some $\beta_H > \beta_L$, some ironed-optimal interim allocation rule X has $\gamma_L > \gamma_H \geq 0$ (Figure 2(a)). In this particular example, observe that $X(\gamma_L,\beta_H) > \lim_{\gamma\to\gamma_L+} X(\gamma,\beta_L)$. Then, for agents with types $\gamma_H < \theta < \gamma_L$, we perform a transport operation that will give us an intermediate rule $\tilde{\tilde{X}}$ (Figure 2(b)). Assume that $H(\beta_H) > H(\beta_L)$. Then, this operation moves a mass equal to $H(\beta_L)$ from agent types $\gamma_H < \theta < \gamma_L$ with outside option β_H to the corresponding agent types with outside option β_L. Thus, $\tilde{\tilde{X}}(\theta,\beta_L) = X(\theta,\beta_H)$ for $\gamma_H < \theta < \gamma_L$. This process also replaces the displaced mass in $X(\theta,\beta_H)$, $\gamma_H < \theta < \gamma_L$, with β_H.

The constructed rule $\tilde{\tilde{X}}(\theta,\beta_H)$, is monotone; however, $\tilde{\tilde{X}}(\theta,\beta_L)$ is not due to a discontinuity at point $\theta = \beta_L$. To fix this problem and construct a rule $\tilde{X}$ with the desired property, we apply a second ironing argument – this time on an allocation rule, that is $\tilde{\tilde{X}}(\theta,\beta_L)$, instead of the virtual

payoff function ψ, to get a monotone rule $\tilde{X}$ (Figure 2(c)). In this construction process, our temporary focus on ironed social welfare is critical: the assumption that $\overline{\psi}$ is nondecreasing is necessary to successfully apply this ironing method on $\tilde{\tilde{X}}$.

We show that the resulting $\tilde{X}$ is incentive-compatible and individually rational, yields at least the same ironed social welfare, and is characterized by increasing participation thresholds. The cases where $H(\beta_H) < H(\beta_L)$ and/or $X(\gamma_L, \beta_H) \leq \lim_{\gamma \to \gamma_L +} X(\gamma, \beta_L)$ are analogous and we present their details in the full proof in Appendix A.

The next lemma builds upon Lemma 5 to show that there exists an ironed-optimal solution that corresponds to a partially symmetric interim allocation rule (class $\mathcal{U}$).

We start from an ironed-optimal rule X, which by Lemma 5 has nondecreasing participation thresholds; in the example of Figure 3(a), we have $\gamma_L < \gamma_H$ for two outside options $\beta_L < \beta_H$. For every interval defined by two consecutive participation thresholds and the endpoints of Θ, we replace the given rule by a new one that replaces the current assignment by the local average of assigned common object types, i.e., $\tilde{\tilde{X}}_a(\theta)$, adjusted by proper weights $H(\beta_i)$. In Figure 3(a), we depict $\tilde{\tilde{X}}_a(\theta)$ in red.

Although $\tilde{\tilde{X}}_a(\theta)$ appears to have the desired form of a partially common interim allocation rule, it might not be incentive-compatible; thus, we apply again an ironing process on function $\tilde{\tilde{X}}_a(\theta)$ to get a nondecreasing function $\hat{X}^* \in \mathcal{S}$ (Figure 3(b)). Then, we use Definition 2 to get the final partially symmetric interim allocation rule X^*. We prove that X^* indeed belongs to class $\mathcal{U}$ and is also ironed-optimal.

Lemma 6. *There exists an ironed-optimal $X^* \in \mathcal{U}$.*

Next, we focus on the interim allocation rule X_{P} corresponding to a positive assortative matching with outside options, which belongs to class $\mathcal{U}$. Recall that, as we defined in (3.1), the rule X_{P} assigns the highest common object types to the highest remaining agent types in a greedy manner.

The rule X_{P} belongs to a general family of greedy rules that we define next. Given any participation thresholds $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \ldots, \gamma_l)$, where $0 = \gamma_0 \leq \gamma_1 \leq \gamma_2 \leq \ldots \leq \gamma_l \leq \gamma_{l+1} = 1$, consider the interim allocation rule $X_{\boldsymbol{\gamma}}$ such that for each $1 \leq i \leq l+1$,

$$\hat{X}_{\boldsymbol{\gamma}}(\theta) = \overline{F}^{-1}\left(\sum_{k=i+1}^{l}\sum_{j=1}^{k} H(\beta_j)\left(\overline{G}(\gamma_k) - \overline{G}(\gamma_{k+1})\right) + \sum_{j=1}^{i} H(\beta_j)\left(\overline{G}(\theta) - \overline{G}(\gamma_{i+1})\right)\right), \gamma_i < \theta \leq \gamma_{i+1}.$$

The rule $X_{\boldsymbol{\gamma}}$ allocates in a greedy manner the same expected object type to all agent types in each interval $(\gamma_i, \gamma_{i+1}]$, $0 < i \leq l$, with outside option at most β_i, for some *given* participation thresholds $\boldsymbol{\gamma}$. Therefore, note that although $X_{\boldsymbol{\gamma}}$ is always incentive-compatible (since it is nondecreasing), it is not necessarily individually rational for all $\boldsymbol{\gamma}$.

Then, $X_{\text{P}}(\theta, \beta)$ is a rule of the class $\mathcal{U}$ of partially symmetric interim allocation rules where $\hat{X}_{\text{P}}$

in (4.10) is defined as

$$\hat{X}_{\mathrm{P}}(\theta) = \hat{X}_{\boldsymbol{\gamma}^{\mathrm{P}}}(\theta)$$

and the thresholds γ_i^{P}, $1 \leq i \leq l$, can be found in Theorem 2. Observe that, by construction, $X_{\mathrm{P}}(\theta, \beta)$ is incentive-compatible and individually rational.

Finally, in the next lemma, we show that X_{P} is ironed-optimal. The proof has two steps. First, making use of Theorem 1, we prove that for any $X \in \mathcal{S}$ with participation thresholds $\boldsymbol{\gamma}$, $X_{\boldsymbol{\gamma}}$ yields weakly higher ironed social welfare than any ironed-optimal X. Second, we show that X_{P} achieves weakly higher ironed social welfare than does any other rule of the form $X_{\boldsymbol{\gamma}}$.

Lemma 7. *The interim allocation rule X_{P} is ironed-optimal, i.e.,*

$$X_{\mathrm{P}} \in \arg\sup_{X \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta).$$

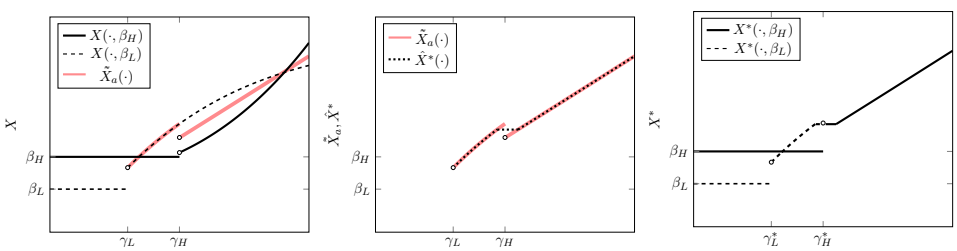


Figure 3: The figure illustrates the local averaging and ironing arguments used in the proof of Lemma 6.

4.4 Social welfare: Optimal interim allocation rule

In the last step of the proof, we construct the interim allocation rule $X_{\mathcal{M}^*}$ from X_{P} (see (3.2)) and prove that it achieves the optimal social welfare.

Lemma 8. *The interim allocation rule $X_{\mathcal{M}^*}$ achieves the optimal social welfare.*

Proof. By Lemma 7, the ironed-optimal interim allocation rule is X_{P}. Given that $\overline{\psi}$ is constant in each interval $[\underline{\theta}_c, \bar{\theta}_c]$, $X_{\mathcal{M}^*}$ also achieves the optimal ironed social welfare.

By (4.6) in Lemma 3, we further get that

$$\begin{aligned}\sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) X_{\mathcal{M}^*}(\theta, \beta_i) \, \mathrm{d}G(\theta) &\leq \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) X_{\mathcal{M}^*}(\theta, \beta_i) \, \mathrm{d}G(\theta) \\ &= \max_{X \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta).\end{aligned}$$

Given that $X_{\mathcal{M}^*}$ is constant in all ironed intervals of ψ, Lemma 3 in turn implies that the previous relation holds with equality and thus $X_{\mathcal{M}^*}$ achieves the optimal social welfare. □

Proof of Theorem 2. Theorem 2 follows as an immediate corollary of Lemma 8. □

5 Downward-private outside options: Proof of Theorem 3

The proof of Theorem 3 is based on two structural results. First, Lemma 9 shows that, conditional on the *truthful* reporting of both agent types and outside options, the expected payoff of an agent of type $\theta \in \Theta$ is increasing in $\beta \in B$, namely $\Phi_{\mathcal{M}^*}(\theta, \beta; \theta, \beta) \geq \Phi_{\mathcal{M}^*}(\theta, \beta'; \theta, \beta')$, for $\beta \geq \beta' \in B$. Second, Lemma 10 establishes an important property of $\mathcal{M}^*$: the payment rule $c_{\mathcal{M}^*}(\cdot, \beta)$ is nonincreasing in $\beta \in B$. This means that, given any two agents of the same type θ but different $\beta' < \beta$, the agent with the *higher* outside option β needs to exert *lower* effort $c_{\mathcal{M}^*}(\theta, \beta) < c_{\mathcal{M}^*}(\theta, \beta')$ to get the same (or weakly higher) expected object type $X_{\mathcal{M}^*}(\theta, \beta) \geq X_{\mathcal{M}^*}(\theta, \beta')$.

Lemma 9. *For any fixed $\theta \in \Theta$, $\Phi_{\mathcal{M}^*}(\theta, \beta; \theta, \beta) \geq \Phi_{\mathcal{M}^*}(\theta, \beta'; \theta, \beta')$, for $\beta \geq \beta' \in B$.*

Proof. By the form of (3.2), it follows that $X_{\mathcal{M}^*}(\theta, \beta) \geq X(\theta, \beta')$ for any $\beta' < \beta \in B$. Thus,

$$\Phi_{\mathcal{M}^*}(\theta, \beta; \theta, \beta) = \theta X(\theta, \beta) - c_{\mathcal{M}^*}(\theta, \beta) = \int_0^\theta X(\theta, \beta) d\theta \geq \int_0^\theta X(\theta, \beta') d\theta = \Phi_{\mathcal{M}^*}(\theta, \beta'; \theta, \beta'),$$

which proves our lemma. □

Lemma 10. *For any fixed $\theta \in \Theta$, $c_{\mathcal{M}^*}(\theta, \beta)$ is nonincreasing in $\beta \in B$.*

The full proof of Lemma 10 can be found in Appendix A. We will use the hypothetical example depicted in Figure 4 to provide some intuition about the proof. We have assumed that $\beta_1 < \omega_1 < \beta_2 < \omega_2 < \omega_3$.

The example refers to a special case where the virtual payoff function $\psi(\theta)$ is nondecreasing. As a result, ψ does not need ironing, which in turn implies that the optimal allocation rule $X_{\mathcal{M}^*}$ coincides with the greedy allocation X_{P}. For the two outside options $\beta_1 < \beta_2$ of our example, we have two different participation thresholds: $\gamma_1^{\mathcal{M}^*} = \gamma_1^{\text{P}}$ and $\gamma_2^{\mathcal{M}^*} = \gamma_2^{\text{P}}$.

As shown on both plots, all agents with types $\theta > \gamma_2^{\mathcal{M}^*}$ receive the same allocation regardless of their outside option. However, agents of the same type $\theta > \gamma_2^{\mathcal{M}^*}$ and different outside options incur different costs. As shown in the right plot, the dark gray area corresponds to the payment $c_{\mathcal{M}^*}(\theta, \beta_2)$ of an agent with high outside option β_2, whereas the sum of the dark and light gray areas corresponds to the payment $c_{\mathcal{M}^*}(\theta, \beta_1)$ of an agent with low outside option β_1. Thus, although both agents receive the same expected object type under $\mathcal{M}^*$, the agent with a *higher* outside option pays a *lower* cost than the other agent, i.e., $c_{\mathcal{M}^*}(\theta, \beta_1) > c_{\mathcal{M}^*}(\theta, \beta_2)$.

This example showcases the main insight in the proof of Lemma 10. However, proving this property for a general, non-monotone ψ is more complicated, since agents with the same type but different outside options might be offered different allocations that potentially pool their outside option with common objects. In the full proof of Lemma 10, which we include in Appendix A, we examine those cases separately.

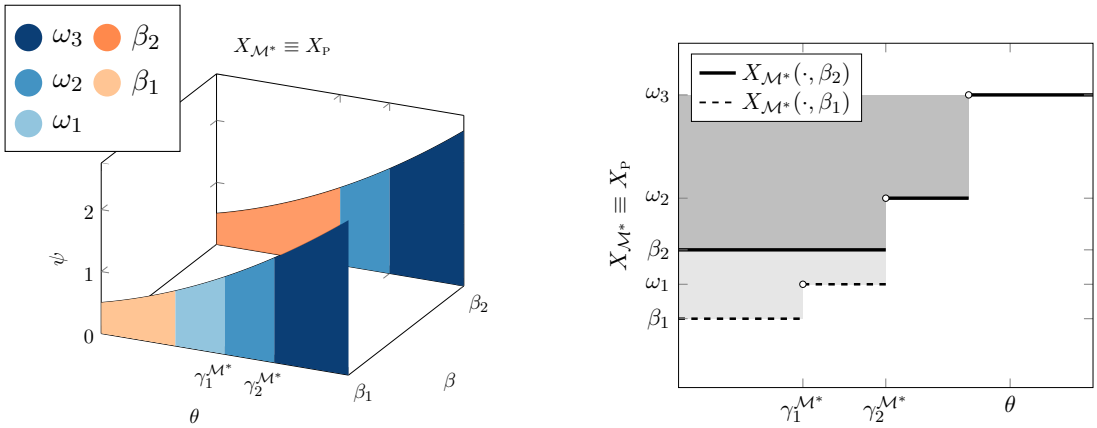


Figure 4: **Left**: In the case of a nondecreasing ψ, the optimal interim allocation rule $X_{\mathcal{M}^*}(\theta, \beta)$ coincides with X_{P}. **Right**: The dashed and solid lines represent $X_{\mathcal{M}^*}(\theta, \beta_1)$ and $X_{\mathcal{M}^*}(\theta, \beta_2)$, respectively. For an agent type θ, the dark gray area corresponds to the payment $c_{\mathcal{M}^*}(\theta, \beta_2)$ of an agent with high outside option β_2, while the sum of the dark and light gray areas corresponds to the payment $c_{\mathcal{M}^*}(\theta, \beta_1)$ of an agent with low outside option β_1. Thus, $c_{\mathcal{M}^*}(\theta, \beta_1) > c_{\mathcal{M}^*}(\theta, \beta_2)$.

Finally, putting everything together, we show that, in our generalized setting with downward-private outside options, the mechanism $\mathcal{M}^*$ is incentive-compatible and remains optimal. Due to the optimality of $\mathcal{M}^*$ for the case of public outside options (Theorem 2), it suffices to prove that no agent has the incentive to misreport her type and outside option under $\mathcal{M}^*$. The key insight is twofold. First, we prove that payments are nonincreasing in β. Second, we show that the effective expected utility of an agent misreporting $\beta' < \beta$ is equal to her utility under reporting her true β.

This also explains why allowing fully private outside options would break the incentive compatibility of $\mathcal{M}^*$. Indeed, we can see this in the example in Figure 4. When two agents of different outside options β_1, β_2 but same agent type θ are offered the same common object quality, the agent with the low outside option β_1 would have the incentive to pretend to be of the high outside option β_2 so that she incurs a lower cost and thus a higher payoff.

Proof of Theorem 3. Due to Theorem 2, it suffices to show that an agent of type θ and outside option β does not have the incentive to misreport his type as $\theta' \neq \theta$ and/or $\beta' < \beta$.

We have that

$$\begin{aligned}\Phi_{\mathcal{M}^*}(\theta,\beta;\theta',\beta') =& \max\{\beta, \theta X_{\mathcal{M}^*}(\theta',\beta) - c_{\mathcal{M}^*}(\theta',\beta')\}\\ \leq& \max\{\beta, \theta X_{\mathcal{M}^*}(\theta',\beta) - c_{\mathcal{M}^*}(\theta',\beta)\}\\ \leq& \theta X_{\mathcal{M}^*}(\theta,\beta) - c_{\mathcal{M}^*}(\theta,\beta)\\ =& \Phi_{\mathcal{M}^*}(\theta,\beta;\theta,\beta).\end{aligned}$$

The intuition behind the first equality is based on the following observation. Consider two agents a, a' with the same type θ and outside options β, β' respectively, where $\beta > \beta'$. Then, by (3.2), it follows that a is assigned by $X_{\mathcal{M}^*}$ to common object types that are a subset of those assigned to a'. In fact, among the common object types assigned to a', those with a type above β are assigned to both a and a', with the same probability.

More formally, the agent of true type θ and option β is offered $X(\theta',\beta')$ when she misreports θ' and $\beta' < \beta$. From the definition of $X_{\mathcal{M}^*} \in S_o$, there exists $x \in \mathcal{D}$ with the following two properties due to the structure of $X_{\mathcal{M}^*}$ in (3.2). First, $x(\theta',\beta,\beta) \geq x(\theta',\beta',\beta') \geq 0$. Second, for all common object types $\omega > \beta$, $x(\theta',\beta,\omega) = x(\theta',\beta',\omega)$. When the agent is offered an object quality $\omega < \beta$, she will always reject it to take β, which implies that the expected object type offered to her is effectively $X(\theta',\beta)$ instead of $X(\theta',\beta')$. Due to the individual rationality constraint, we must also consider the maximum between β and $\theta X_{\mathcal{M}^*}(\theta',\beta) - c_{\mathcal{M}^*}(\theta',\beta')$ as her final payoff $\Phi_{\mathcal{M}^*}(\theta,\beta;\theta',\beta')$ from misreporting.

The first inequality arises from Lemma 10. The second inequality follows from the incentive compatibility and individual rationality of $\mathcal{M}^*$. Therefore, $\mathcal{M}^*$ is incentive-compatible and individually rational in our setting with downward-private outside options, and by Theorem 2 achieves the optimal social welfare. □

6 Conclusion

The COVID-19 pandemic has brought new challenges for rationing scarce resources including including testing, vaccinations and other medical supplies. Naturally ethical and cost-benefit principles guide how to design priorities (Emanuel et al., 2020; Dixon and Welch, 1991) in the allocation of such scarce resources. Yet, the social planner should further account for agents' private information regarding their taste and outside options. We resolve this mechanism design problem in an environment with vertically differentiated objects.

One followup research study arises from allowing agents to also inflate their outside options. Another direction is the design of mechanisms when agents have type-dependent outside options and arbitrary preferences over objects.

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A Omitted Proofs

A.1 Proof of Lemma 2

Lemma 2. *Let $\mathcal{M}$ be an incentive-compatible and individually rational mechanism $\mathcal{M}$. Then, $X_{\mathcal{M}}(\theta, \beta_i)$ and $c_{\mathcal{M}}(\theta, \beta_i)$ satisfy the following properties:*

(i) for each fixed $\beta_i \in B$, $X_{\mathcal{M}}(\theta, \beta_i)$ is nondecreasing in $\theta \in \Theta$; and

(ii) for all $\theta \in \Theta$, $\beta_i \in B$,

$$c_{\mathcal{M}}(\theta, \beta_i) = \theta X_{\mathcal{M}}(\theta, \beta_i) - \int_0^\theta X_{\mathcal{M}}(z, \beta_i)\,\mathrm{d}z. \tag{4.1}$$

Furthermore, for each $1 \le i \le l$, there exists a participation threshold $\gamma_i^{\mathcal{M}}$ such that $X_{\mathcal{M}}(\theta, \beta_i) > \beta_i$ for all $\theta > \gamma_i^{\mathcal{M}}$ and $X_{\mathcal{M}}(\theta, \beta_i) = \beta_i$ for all $\theta \le \gamma_i^{\mathcal{M}}$.

Proof. Fix any $1 \le i \le l$. Let $D_i^{\mathcal{M}}$ be the set of agent types θ with outside option β_i such that $\Phi_{\mathcal{M}}(\theta, \theta, \beta_i) > \theta\beta_i$; thus, by Lemma 1, $X_{\mathcal{M}}(\theta, \beta_i) = \beta_i$ and $c_{\mathcal{M}}(\theta, \beta_i) = 0$ for all $\theta \notin D_i^{\mathcal{M}}$.

We will first prove part (i) and then show that $D_i^{\mathcal{M}} = (\gamma_i^{\mathcal{M}}, 1]$ for some $\gamma_i^{\mathcal{M}} \in \Theta$. Let

$$\gamma_i^{\mathcal{M}} = \inf\{\gamma \in \Theta : \gamma\beta_i < \gamma X_{\mathcal{M}}(\gamma, \beta_i) - c_{\mathcal{M}}(\gamma, \beta_i)\}.$$

By the incentive compatibility constraint and for $\theta' > \theta$, it follows that

$$\theta(X_{\mathcal{M}}(\theta, \beta_i) - X_{\mathcal{M}}(\theta', \beta_i)) \ge t_{\mathcal{M}}(\theta, \beta_i) - t_{\mathcal{M}}(\theta', \beta_i) \ge \theta'(X_{\mathcal{M}}(\theta, \beta_i) - X_{\mathcal{M}}(\theta', \beta_i))$$

which in turn implies that $X_{\mathcal{M}}(\theta, \beta_i)$ must be nondecreasing in θ for fixed β_i (part (i)).

As a result of part (i), $X_{\mathcal{M}}(\theta, \beta_i) \le \beta_i$ if and only if $\theta \le \gamma_i^{\mathcal{M}}$ for some $\gamma_i^{\mathcal{M}} \in \Theta$. An agent of type $\gamma_i^{\mathcal{M}}$ and outside option β_i is indifferent with payoff $\gamma_i^{\mathcal{M}}\beta_i = \gamma_i^{\mathcal{M}} X_{\mathcal{M}}(\gamma^{\mathcal{M}}, \beta_i) - c_{\mathcal{M}}(\gamma_i^{\mathcal{M}}, \beta_i)$ and – by our convention – always chooses β_i. Therefore, $D_i^{\mathcal{M}} = (\gamma_i^{\mathcal{M}}, 1]$.

Finally, the proof of eq. (4.1) follows as in Myerson (1981). □

A.2 Proof of Lemma 3

Lemma 3. *For any $X \in \mathcal{S}$ and $\beta_i \in B$, it holds that*

$$\int_0^1 \psi(\theta) X(\theta, \beta_i)\,\mathrm{d}G(\theta) \le \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i)\,\mathrm{d}G(\theta). \tag{4.6}$$

If $X(\theta, \beta_i)$ is constant in each ironed interval of ψ, then

$$\int_0^1 \psi(\theta, \beta_i) X(\theta, \beta_i)\,\mathrm{d}G(\theta) = \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i)\,\mathrm{d}G(\theta). \tag{4.7}$$

Conversely, if for some $X \in \mathcal{S}$ it holds that

$$\sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta) = \max_{Y \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) Y(\theta, \beta_i) \, \mathrm{d}G(\theta), \tag{4.8}$$

then for all $\beta_i \in B$, $X(\theta, \beta_i)$ is constant in any ironed interval of ψ.

Proof. By relation (4.5), we have that

$$\begin{aligned}
&\int_0^1 \psi(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta) - \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta) = \int_0^1 (K'(G(\theta)) - J'(G(\theta))) X(\theta, \beta_i) \, \mathrm{d}G(\theta) \\
=& X(1, \beta_i)(H(G(1)) - J(G(1))) - X(0, \beta_i)(K(G(0)) - J(G(0))) - \int_0^1 (K(G(\theta)) - J(G(\theta))) \, \mathrm{d}X(\theta, \beta_i) \\
=& - \int_0^1 (K(G(\theta)) - J(G(\theta))) \, \mathrm{d}X(\theta, \beta_i) \leq 0
\end{aligned} \tag{A.1}$$

since by construction J and K must coincide on the endpoints of $[0, 1]$, $J \leq K$, and the measure induced by X is nonnegative since X is nondecreasing by Lemma 2.

It is straightforward to see that, if $X(\theta, \beta_i)$ is constant whenever $J(G(\theta)) < H(G(\theta))$, then the difference in (A.1) is exactly 0.

Finally, we prove the last statement by contradiction. Let (4.8) hold for some X^*, thus, due to eq. (4.6), it also holds that

$$X^* \in \arg\max_{Y \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) Y(\theta, \beta_i) \, \mathrm{d}G(\theta).$$

However, assume that there is some ironed interval $I = [\theta_1, \theta_2] \subset \Theta$, that is $J(G(\theta)) < K(G(\theta))$ for all $\theta \in I$ and $X^*(\theta_1) < X^*(\theta_2)$. Then, for each $\beta_i \in B$, we define $Z \in \mathcal{S}$ as

$$Z(\theta, \beta_i) = \begin{cases} X^*(\theta, \beta_i) & \theta \in \Theta \setminus I \\ \frac{\int_{\theta_1}^{\theta_2} X^*(\theta, \beta_i) \, \mathrm{d}G(\theta)}{G(\theta_2) - G(\theta_1)} & \theta \in I. \end{cases} \tag{A.2}$$

The rule Z is feasible, that is $Z \in \mathcal{S}_o$, as it reallocates exactly the same objects as X^*; specifically, it is identical to X^* except for agent types in the interval I where it randomly allocates the same object types to all $\theta \in I$. Furthermore, Z preserves the monotonicity of X^* on Θ since Z is constant on I, i.e., $Z(\theta, \beta_i) = \mathbb{E}[X^*(\theta, \beta_i) \mid \theta_1 \leq \theta \leq \theta_2]$, $\forall \theta \in I$, and $X^*(\theta_1, \beta_i) \leq \mathbb{E}[X^*(\theta, \beta_i) \mid \theta_1 \leq \theta \leq \theta_2] \leq X^*(\theta_2, \beta_i)$.

By (A.1), summing over all $\beta_i \in B$, we get that

$$\begin{aligned}&\sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta)(X^*(\theta,\beta_i) - Z(\theta,\beta_i))\,\mathrm{d}G(\theta)\\ =&\sum_{i=1}^{l} H(\beta_i) \left[\int_I \overline{\psi}(\theta)(X^*(\theta,\beta_i) - Z(\theta,\beta_i))\,\mathrm{d}G(\theta) - \int_I (K(G(\theta)) - J(G(\theta)))\,\mathrm{d}X^*(\theta,\beta_i)\right].\end{aligned} \tag{A.3}$$

Since the measure induced by X^* is strictly positive for all or part of interval I and $K(G(\theta)) > J(G(\theta))$ for $\theta \in I$, each term $-\int_I (H(G(\theta)) - J(G(\theta)))(X^*)'(\theta,\beta_i)\,\mathrm{d}G(\theta)$ is strictly negative. For the first term, it holds that $\int_I \overline{\psi}(\theta)(X^*(\theta,\beta_i) - Z(\theta,\beta_i))\,\mathrm{d}G(\theta) = 0$, since by Definition 1, J is linear on I and therefore $\psi(\theta) = J'(G(\theta))$ is constant on I. Thus, we get

$$\sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) X^*(\theta,\beta_i)\,\mathrm{d}G(\theta) < \sum_{i=1}^{l} H(\beta_i) \int_0^1 \psi(\theta) Z(\theta,\beta_i)\,\mathrm{d}G(\theta),$$

which is a contradiction, since X^* was assumed to achieve the optimal social welfare. This completes the proof of the lemma. □

A.3 Proof of Lemma 5

Lemma 5. *Let $X \in \mathcal{S}$ be ironed-optimal. Then, there exists a $\tilde{X} \in \mathcal{S}$ with nondecreasing participation thresholds $\gamma_1 \leq \gamma_2 \leq \ldots \leq \gamma_l$, which yields weakly higher ironed social welfare than X.*

Proof. The proof is by contradiction. Assume that for some $\beta_H > \beta_L$, the interim allocation rule X has $\gamma_L > \gamma_H \geq 0$. We consider the following two cases.

<u>*Case (i): $X(\gamma_L,\beta_H) \leq \lim_{\gamma\to\gamma_L+} X(\gamma,\beta_L)$.*</u> We construct a new rule $\tilde{X}$ from X as follows; for agents with outside option β_H, we define

$$\tilde{X}(\theta,\beta_H) = \begin{cases} X(\theta,\beta_H), & \theta \geq \gamma_L,\\ \mathbf{1}_{\{H(\beta_H)>H(\beta_L)\}} \frac{H(\beta_H)-H(\beta_L)}{H(\beta_H)} X(\theta,\beta_H) + \frac{\min\{H(\beta_H),H(\beta_L)\}}{H(\beta_H)}\beta_H, & \theta < \gamma_L.\end{cases} \tag{A.4}$$

Similarly, for agents with β_L, we define

$$\tilde{X}(\theta,\beta_L) = \begin{cases} X(\theta,\beta_L), & \theta \geq \gamma_L,\\ \mathbf{1}_{\{H(\beta_H)<H(\beta_L)\}} \frac{H(\beta_L)-H(\beta_H)}{H(\beta_L)} \beta_L + \frac{\min\{H(\beta_H),H(\beta_L)\}}{H(\beta_L)} X(\theta,\beta_H), & \gamma_H < \theta < \gamma_L,\\ \beta_L, & \theta \leq \gamma_H.\end{cases} \tag{A.5}$$

Lemma 4 guarantees that such a construction (A.4) - (A.5) is feasible as $X(\theta,\beta_H)$ does not assign β_H to any agent of type $\theta \geq \gamma_H$.

The new rule $\hat{X}$ is individually rational since by construction $\hat{X}(\theta, \beta) \geq \beta$ for both $\beta \in \{\beta_H, \beta_L\}$.

We also show that the rule $\hat{X}$ is incentive-compatible. We begin with $\tilde{X}(\theta, \beta_H)$. In each of the intervals $[0, \gamma_L)$ and $[\gamma_L, 1]$, $\tilde{X}(\theta, \beta_H)$ is nondecreasing as a linear combination of the nondecreasing rule $X(\theta, \beta_H)$ and constant β_H. At the discontinuity point $\theta = \gamma_L$, since $X(\theta, \beta_H) \geq \beta_H$ for all $\theta > \gamma_H$, definition (A.4) implies that $\tilde{X}(\gamma_L, \beta_H) = X(\gamma_L, \beta_H) \geq \tilde{X}(\theta, \beta_H)$ for all $\theta < \gamma_L$. Consequently, $\tilde{X}(\theta, \beta_H)$ is nondecreasing in any $\theta \in \Theta$ and thus, due to Lemma 2, incentive-compatible.

A similar argument shows that $\tilde{X}(\theta, \beta_L)$ is also incentive-compatible. Observe that at the highest discontinuity point $\theta = \gamma_L$, $\tilde{X}(\gamma_L, \beta_L) = X(\gamma_L, \beta_L) \geq X(\gamma_L, \beta_H)$. Regarding the lowest discontinuity point $\theta = \gamma_H$, we have that, for $\gamma_L > \theta > \gamma_H$, definition (A.5) together with the fact that $X(\theta, \beta_H) \geq \beta_H > \beta_L$ implies that $\tilde{X}(\theta, \beta_L) > \beta_L$.

Finally, by construction, it follows that the ironed social welfare remains the same under both rules X and $\tilde{X}$ as it can be shown through a direct substitution of the definition of $\tilde{X}$ into our objective.

Case (ii): $X(\gamma_L, \beta_H) > \lim_{\gamma \to \gamma_L +} X(\gamma, \beta_L)$. We construct $\tilde{X}$ as follows. For agents with outside option β_H, we define $\tilde{X}(\theta, \beta_H)$ as in (A.4). As in case (i), $\tilde{X}(\theta, \beta_H)$ is incentive-compatible and individually rational.

Next, we construct $\tilde{X}(\theta, \beta_L)$ in two steps. First, we define

$$\tilde{\tilde{X}}(\theta, \beta_L) = \begin{cases} X(\theta, \beta_L), & \theta > \gamma_L, \\ \mathbf{1}_{\{H(\beta_H) < H(\beta_L)\}} \frac{H(\beta_L) - H(\beta_H)}{H(\beta_L)} \beta_L + \frac{\min\{H(\beta_H), H(\beta_L)\}}{H(\beta_L)} X(\theta, \beta_H), & \gamma_H \leq \theta \leq \gamma_L, \\ \beta_L, & \theta \leq \gamma_H. \end{cases} \tag{A.6}$$

Due to our assumption $X(\gamma_L, \beta_H) > \lim_{\gamma \to \gamma_L +} X(\gamma, \beta_L)$ and the above first step of construction, $\tilde{\tilde{X}}(\theta, \beta_L)$ may not be nondecreasing across all Θ. Thus, we move to the second step of our construction.

We follow a convexification process. Define

$$\tilde{K}(\theta) = \int_0^\theta \tilde{\tilde{X}}(G^{-1}(\tau), \beta_L) \, \mathrm{d}\tau,$$

and $\tilde{J}(\theta)$ to be the convex hull of $\tilde{K}$. Let $[a, b] \subseteq [\gamma_H, 1]$ be the (unique) maximal interval such that $\tilde{J}(\theta) < \tilde{K}(\theta)$.[12] Then, define $\tilde{X}(\theta, \beta_L) = \tilde{J}'(G(\theta))$ and extend the definition of J', if necessary, by right continuity.

By construction, $\tilde{X}(\theta, \beta_L)$ is nondecreasing. Since $X(\gamma_H, \beta_H) \geq \beta_H > \beta_L$, it also holds that $\tilde{X}(\theta, \beta_L) \geq \beta_L$. Thus, $\tilde{X}(\theta, \beta_L)$ is incentive-compatible.

[12] This interval is unique because $\hat{X}_1$ is nondecreasing except in a neighborhood of γ_L.

Finally, note that, as in case (i), the ironed social welfare under $\tilde{X}(\theta, \beta_H)$ and $\tilde{\tilde{X}}(\theta, \beta_L)$ remains the same as under X. Therefore, the total change in the ironed social welfare, from changing rule X to $\tilde{X}$, is

$$\begin{aligned}
&H(\beta_L) \int_a^b \overline{\psi}(\theta) \left(\tilde{X}(\theta, \beta_L) - \tilde{\tilde{X}}(\theta, \beta_L) \right) \mathrm{d}G(\theta) \\
=&H(\beta_L) \int_a^b \overline{\psi}(\theta) \left(\tilde{J}'(G(\theta)) - \tilde{K}'(G(\theta)) \right) \mathrm{d}\theta \\
=&H(\beta_L) \left(\overline{\psi}(b) \left(\tilde{J}(G(b)) - \tilde{K}(G(b)) \right) - \overline{\psi}(a) \left(\tilde{J}(G(a)) - \tilde{K}(G(a)) \right) \right) \\
&- H(\beta_L) \int_a^b \overline{\psi}'(\theta) \left(\tilde{J}(G(\theta)) - \tilde{K}(G(\theta)) \right) \mathrm{d}\theta \\
=&- H(\beta_L) \int_a^b \overline{\psi}'(\theta) \left(\tilde{J}(G(\theta)) - \tilde{K}(G(\theta)) \right) \mathrm{d}\theta \geq 0,
\end{aligned}$$

where the third equality follows from the fact that $\tilde{J}$ and $\tilde{K}$ coincide at the ends of an ironed interval $[a, b]$, and the inequality follows from the facts that $\tilde{J}(G(\theta)) < \tilde{K}(G(\theta))$, $\theta \in (a, b)$ and $\overline{\psi}' \geq 0$. Thus, the ironed social welfare weakly improves under $\tilde{X}$.

To conclude the proof, we showed that in both cases (i) and (ii), given any rule ironed-optimal $X \in \mathcal{S}$ with $\gamma_H < \gamma_L$, there exists a rule $\tilde{X} \in \mathcal{S}$ with participation thresholds $\tilde{\gamma}_H = \gamma_L \geq \tilde{\gamma}_L = \gamma_H$ that yields weakly higher ironed social welfare. Repeating this process for all participation thresholds, we conclude that there exists an ironed-optimal interim allocation rule with nondecreasing participation thresholds. □

A.4 Proof of Lemma 6

Lemma 6. *There exists an ironed-optimal $X^* \in \mathcal{U}$.*

Proof. Let $X \in \mathcal{S}$ be an ironed-optimal interim allocation rule. Without loss of generality, as guaranteed by Lemma 5, we assume that X has nondecreasing participation thresholds $\gamma_1 \leq \gamma_2 \leq \ldots \leq \gamma_l$.

Given X, we construct a new rule $\tilde{\tilde{X}} \in \mathcal{U}$. Specifically, for each $\gamma_{i-1} \leq \theta < \gamma_i$, where $0 < i \leq l+1$, $\gamma_0 = 0$, $\gamma_{l+1} = 1$, let $\tilde{\tilde{X}}_a(\theta)$ be the average common object type allocated to all agents θ who are not assigned to their outside option by X, i.e.,

$$\tilde{\tilde{X}}_a(\theta) = \frac{\sum_{j=1}^{i-1} H(\beta_j) X(\theta, \beta_j)}{\sum_{j=1}^{i-1} H(\beta_j)},$$

and define

$$\tilde{\tilde{X}}(\theta, \beta_i) = \begin{cases} \tilde{\tilde{X}}_a(\theta), & \theta > \gamma_i \\ \beta_i, & \theta \leq \gamma_i. \end{cases}$$

By construction, $\tilde{\tilde{X}}$ yields equal ironed social welfare to X. However, $\tilde{\tilde{X}}$ may not be incentive-compatible.

To construct an incentive-compatible and individually rational rule $X \in \mathcal{U}$, we apply to $\tilde{\tilde{X}}_a(\theta)$ an ironing process (similar to case (ii) in the proof of Lemma 5). Specifically, let

$$\tilde{G}(\theta) = \frac{\sum_{j=1}^{i-1} H(\beta_j) G(\theta)}{\sum_{i=1}^{l+1} (G(\gamma_i^{\mathrm{P}}) - G(\gamma_{i-1}^{\mathrm{P}})) \sum_{j=1}^{i} H(\beta_j)}, \gamma_{i-1} \leq \theta < \gamma_i;$$

$\tilde{G}$ corresponds to the distribution of agent types if we ignore all agents preferring their outside options under X^*. (For convenience we introduced the notation $H(\beta_{l+1}) = 0$.)

Then, define

$$\tilde{K}(\theta) = \int_0^\theta \tilde{\tilde{X}}_a(\tilde{G}^{-1}(\tau))\, \mathrm{d}\tau,$$

and let $\tilde{J}(\theta)$ be the convex hull of $\tilde{K}$. Next, define $\hat{X}^*(\theta) = \tilde{J}'(\tilde{G}(\theta))$ and extend the definition of $\tilde{J}'$, if necessary, by right continuity. Then, let $\gamma_i^* = \arg\inf_{\theta \geq \gamma_i} \{\hat{X}(\theta) > \beta_i\}$ and

$$X^*(\theta, \beta_i) = \begin{cases} \hat{X}^*(\theta), & \theta > \gamma_i^* \\ \beta_i, & \theta \leq \gamma_i^*. \end{cases}$$

By construction, X^* belongs to class $\mathcal{U}$, is individually rational and incentive-compatible.

Finally, we show that the ironed social welfare weakly improves under X^* compared to X. Consider the collection of maximal intervals, namely (a_m, b_m) indexed by $m \in C$, such that $\tilde{J}(\tilde{G}(\theta)) < \tilde{K}(\tilde{G}(\theta))$. Given that $\tilde{\tilde{X}}$ yields equal ironed social welfare to X, the total change in the ironed social welfare, from changing rule X to $\tilde{\tilde{X}}$, is bounded below by

$$\begin{aligned}
&\sum_{m \in C} \sum_{i=1}^{l+1} (G(\gamma_i) - G(\gamma_{i-1})) \sum_{j=1}^{i} H(\beta_j) \int_{a_m}^{b_m} \overline{\psi}(\theta) \left(\hat{X}^*(\theta) - \tilde{\tilde{X}}_a(\theta) \right) \mathrm{d}\tilde{G}(\theta) \\
= &\sum_{m \in C} \sum_{i=1}^{l+1} (G(\gamma_i) - G(\gamma_{i-1})) \sum_{j=1}^{i} H(\beta_j) \int_{a_m}^{b_m} \overline{\psi}(\theta) \left(\tilde{J}'(\tilde{G}(\theta)) - \tilde{K}'(\tilde{G}(\theta)) \right) \mathrm{d}\tilde{G}(\theta) \\
= &\sum_{m \in C} \sum_{i=1}^{l+1} (G(\gamma_i) - G(\gamma_{i-1})) \sum_{j=1}^{i} H(\beta_j) \Bigg[\overline{\psi}(b_m) \left(\tilde{J}(\tilde{G}(b_m)) - \tilde{K}(\tilde{G}(b_m)) \right) \\
&- \overline{\psi}(a_m) \left(\tilde{J}(\tilde{G}(a_m)) - \tilde{K}(\tilde{G}(a_m)) \right) - \int_{a_m}^{b_m} \overline{\psi}'(\theta) \left(\tilde{J}(\tilde{G}(\theta)) - \tilde{K}(\tilde{G}(\theta)) \right) \mathrm{d}\theta \Bigg] \\
= &- \sum_{m \in C} \sum_{i=1}^{l+1} (G(\gamma_i) - G(\gamma_{i-1})) \sum_{j=1}^{i} H(\beta_j) \int_{a_m}^{b_m} \overline{\psi}'(\theta) \left(\tilde{J}(\tilde{G}(\theta)) - \tilde{K}(\tilde{G}(\theta)) \right) \mathrm{d}\theta \geq 0,
\end{aligned}$$

where the third equality follows from the fact that $\tilde{J}$ and $\tilde{K}$ coincide at the ends of an ironed interval $[a_m, b_m]$, and the inequality follows from the facts that $\tilde{J}(\tilde{G}(\theta)) < \tilde{K}(\tilde{G}(\theta))$, $\theta \in (a_m, b_m)$,

and $\overline{\psi}' \geq 0$. Thus, the ironed social welfare weakly improves under X^* implying that X^* is also optimal.

□

A.5 Proof of Lemma 7

Lemma 7. *The interim allocation rule X_{P} is ironed-optimal, i.e.,*

$$X_{\mathrm{P}} \in \arg\sup_{X \in \mathcal{S}} \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta).$$

Proof. Suppose some $X \in S$ is optimal.

Step 1: *We show that for any $X \in \mathcal{S}$ with participation thresholds $\boldsymbol{\gamma}$, $X_{\boldsymbol{\gamma}}$ yields weakly higher ironed social welfare than X.* Indeed, following a reasoning similar to Lemma 4, Theorem 1 implies that

$$\sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) X(\theta, \beta_i) \, \mathrm{d}G(\theta) \leq \sum_{i=1}^{l} H(\beta_i) \int_0^1 \overline{\psi}(\theta) X_{\boldsymbol{\gamma}}(\theta, \beta_i) \, \mathrm{d}G(\theta), \tag{A.7}$$

where $X_{\boldsymbol{\gamma}}$ is defined given the known and fixed participation thresholds $\boldsymbol{\gamma}$ of X.

Step 2: *We show that X_{P} achieves weakly higher ironed social welfare than $X_{\boldsymbol{\gamma}}$.* If $\gamma_i = \gamma_i^{\mathrm{P}}$, for all $1 \leq i \leq l$, then $X_{\boldsymbol{\gamma}} \equiv X_{\mathrm{P}}$ and the proof is complete. Assume now that $\boldsymbol{\gamma}$ differ from X_{P}'s participation thresholds $\boldsymbol{\gamma}$ at index i for the first time (while $\gamma_j = \gamma_j^{\mathrm{P}}$ for all $j > i$).

Consider $X_{\boldsymbol{\gamma}'}$, where $\boldsymbol{\gamma}' = (\gamma_1, \ldots, \gamma_{i-1}, \gamma_i^{\mathrm{P}}, \ldots, \gamma_l^{\mathrm{P}})$. We study the following two cases.

Case (i): $\gamma_i < \gamma_i^{\mathrm{P}}$. By the definition of γ_i^{P}, we have that, given participation thresholds $\gamma_{i+1}^{\mathrm{P}}$ to γ_l^{P}, γ_i^{P} is the lowest participation threshold such that the allocation rule is incentive-compatible for agents with outside option β_i, which means that under $X_{\boldsymbol{\gamma}}(\theta, \beta_i)$ agents with types $\theta < \gamma_i^{\mathrm{P}}$ choose β_i to $X_{\boldsymbol{\gamma}}(\theta, \beta_i)$ anyway. Consequently, $X_{\boldsymbol{\gamma}}(\theta, \beta_i)$ and $X_{\boldsymbol{\gamma}'}(\theta, \beta_i)$ achieve the same ironed social welfare for agents with β_i.

In contrast to $X_{\boldsymbol{\gamma}}(\theta, \beta_i)$, $X_{\boldsymbol{\gamma}'}(\theta, \beta_i)$ allocates the previously wasted objects $X_{\boldsymbol{\gamma}}(\theta, \beta_i)$, $\gamma_i \leq \theta < \gamma_i^{\mathrm{P}}$, to agents with lower outside options β_j, $j < i$. Thus, the ironed social welfare increases.

Case (ii): $\gamma_i > \gamma_i^{\mathrm{P}}$. In this case, $X_{\boldsymbol{\gamma}'}$ achieves higher ironed social welfare than $X_{\boldsymbol{\gamma}}$ since all object types initially assigned to agents $\theta < \gamma_i$ are now assigned to weakly higher agent types and, as a result, to weakly higher values of $\overline{\psi}$.

In both cases, $X_{\boldsymbol{\gamma}'}$ yields strictly higher social welfare than $X_{\boldsymbol{\gamma}}$ and – due to (A.7) – higher than X as well. This contradicts the optimality of X. Thus, X_{P} is optimal. □

A.6 Proof of Lemma 10

Lemma 10. *For any fixed $\theta \in \Theta$, $c_{\mathcal{M}^*}(\theta, \beta)$ is nonincreasing in $\beta \in B$.*

Proof. Due to Theorem 2, $c_{\mathcal{M}^*}(\theta, \beta_i)$ is piece-wise constant for each $\beta_i \in B$. Specifically, let Σ_i denote the set of all discontinuity points of $c_{\mathcal{M}^*}(\cdot, \beta_i)$; let σ_j denote the j-th largest point in Σ_i.

Thus, (3.2)-(3.3) implies that for any two consecutive $\sigma_j, \sigma_{j+1} \in \Sigma_i$ and $\beta_i \in B$, it holds

$$c_{\mathcal{M}^*}(\theta, \beta_i) = c_{\mathcal{M}^*}(\sigma_{j+1}, \beta_i) = \lim_{\theta \to \sigma_j +} c_{\mathcal{M}^*}(\theta, \beta_i),$$

where $\theta \in (\sigma_j, \sigma_{j+1}]$.

Consequently, to prove our claim, it suffices to show that for any given $1 < i \leq l$ and all $\sigma \in \Sigma_i$,

$$\lim_{\theta \to \sigma^+} c_{\mathcal{M}^*}(\theta, \beta_{i-1}) \geq \lim_{\theta \to \sigma^+} c_{\mathcal{M}^*}(\theta, \beta_i).$$

Step 1: *We show that, for $\sigma = \gamma_i \in \Sigma_i$, $\lim_{\theta \to \gamma_i^+} c_{\mathcal{M}^*}(\theta, \beta_{i-1}) \geq \lim_{\theta \to \gamma_i^+} c_{\mathcal{M}^*}(\theta, \beta_i)$.* We have the following cases: either $\gamma_i = \underline{\theta}_c$ for some $c \in \mathcal{C}$, or $\gamma_i \notin \cup_{c \in \mathcal{C}} [\underline{\theta}_c, \overline{\theta}_c]$.

Case (i): $\gamma_i = \underline{\theta}_c$ for some $c \in \mathcal{C}$, i.e., γ_i coincides with the left endpoint of an ironed interval of ψ. Since $X_{\mathcal{M}^*} \in \mathcal{S}$, there exists $x \in \mathcal{D}$, such that $x(\theta, \beta_i, \beta_i) \geq x(\theta, \beta_{i-1}, \beta_{i-1}) \geq 0$, where $\theta \in (\underline{\theta}_c, \overline{\theta}_c]$; due to the structure of $X_{\mathcal{M}^*}$, for all common object types $\omega > \beta_i$, it also holds that $x(\theta, \beta_{i-1}, \omega) = x(\theta, \beta_i, \omega)$. For clarity of exposition consider that $x(\theta, \beta_{i-1}, \beta_{i-1}) = 0$; the case where $x(\theta, \beta_{i-1}, \beta_{i-1}) > 0$ follows using an identical argument.

For any $\theta \in (\underline{\theta}_c, \overline{\theta}_c]$, we have that

$$\begin{aligned}
&\lim_{\theta \to \gamma_i^+} c_{\mathcal{M}^*}(\theta, \beta_{i-1}) - \lim_{\theta \to \gamma_i^+} c_{\mathcal{M}^*}(\theta, \beta_i) \\
=&\gamma_i \left[\sum_{\omega \in \Omega : \omega \leq \beta_i} \omega x(\theta, \beta_{i-1}, \omega) + \sum_{\omega \in \Omega : \omega > \beta_i} \omega x(\theta, \beta_{i-1}, \omega) \right] - (\gamma_i X_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}) - c_{\mathcal{M}^*}(\gamma_i, \beta_{i-1})) \\
&- \gamma_i \left[\beta_i \sum_{\omega \in \Omega : \omega \leq \beta_i} x(\theta, \beta_{i-1}, \omega) + \sum_{\omega \in \Omega : \omega > \beta_i} \omega x(\theta, \beta_{i-1}, \omega) \right] + \gamma_i \beta_i \\
=&\gamma_i \left[\sum_{\omega \in \Omega : \omega \leq \beta_i} \omega x(\theta, \beta_{i-1}, \omega) + \beta_i \sum_{\omega \in \Omega : \omega > \beta_i} x(\theta, \beta_{i-1}, \omega) \right] - \gamma_i X_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}) + c_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}) \geq 0,
\end{aligned}$$

where the inequality follows from the fact that $X_{\mathcal{M}^*}(\theta, \beta_i)$ is nondecreasing thus all the objects that $X_{\mathcal{M}^*}(\gamma_i, \beta_{i-1})$ allocates have weakly lower quality than any object that $X_{\mathcal{M}^*}(\theta, \beta_{i-1})$ as well

as β_i, therefore

$$X_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}) < \sum_{\omega \in \Omega : \omega \leq \beta_i} \omega x(\theta, \beta_{i-1}, \omega) + \beta_i \sum_{\omega \in \Omega : \omega > \beta_i} \omega x(\theta, \beta_{i-1}, \omega).$$

Case (ii): $\gamma_i \notin \cup_{c \in \mathcal{C}} [\underline{\theta}_c, \overline{\theta}_c]$. In this case, $X_{\mathcal{M}^*}(\theta, \beta_i) = X_{\mathcal{M}^*}(\theta, \beta_{i-1})$ for all $\theta \in (\gamma_i, \sigma^*]$, where $\sigma^* \triangleq \min\{\sigma \in \Sigma_i \cup \{1\} : \sigma > \gamma_i\}$.

For any $\theta \in (\gamma_i, \sigma^*]$, we have that

$$\begin{aligned}
&\lim_{\theta \to \gamma_i^+} c_{\mathcal{M}^*}(\theta, \beta_{i-1}) - \lim_{\theta \to \gamma_i^+} c_{\mathcal{M}^*}(\theta, \beta_i) \\
=&\gamma_i X_{\mathcal{M}^*}(\theta, \beta_{i-1}) - (\gamma_i X_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}) - c_{\mathcal{M}^*}(\gamma_i, \beta_{i-1})) - (\gamma_i X_{\mathcal{M}^*}(\theta, \beta_{i-1}) - \gamma_i \beta_i) \\
=&\gamma_i \beta_i - \gamma_i X_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}) + c_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}) \\
=&\gamma_i \beta_i - \Phi_{\mathcal{M}^*}(\gamma_i, \beta_{i-1}; \gamma_i, \beta_{i-1}) \\
\geq&\gamma_i \beta_i - \Phi_{\mathcal{M}^*}(\gamma_i, \beta_i; \gamma_i, \beta_i) = 0,
\end{aligned}$$

where the inequality follows from Lemma 9 and the last equality follows from the definition of the participation threshold γ_i.

Step 2: *We show that for any $\sigma_j \in S_i$, $\lim_{\theta \to \sigma_j^+} c_{\mathcal{M}^*}(\theta, \beta_{i-1}) \geq \lim_{\theta \to \sigma_j^+} c_{\mathcal{M}^*}(\theta, \beta_i)$.* Assume that $\sigma_j \neq \gamma_i, 0$. We considered the case of $\sigma_j = \gamma_i$ in Step 1, while for $\sigma_j = 0$, we trivially have $c_{\mathcal{M}^*}(0, \beta) = 0$, for all $\beta \in B$.

For any $\theta \in (\sigma_j, \sigma_{j+1}]$, it holds that $X_{\mathcal{M}^*}(\theta, \beta_i) = X_{\mathcal{M}^*}(\theta, \beta_{i-1})$. Consequently,

$$\begin{aligned}
&\lim_{\theta \to \sigma_j{}^+} c_{\mathcal{M}^*}(\theta, \beta_{i-1}) - \lim_{\theta \to \sigma_j{}^+} c_{\mathcal{M}^*}(\theta, \beta_i) \\
=&\sigma_j X_{\mathcal{M}^*}(\theta, \beta_{i-1}) - \Phi_{\mathcal{M}^*}(\sigma, \beta_{i-1}; \sigma_k, \beta_{i-1}) - (\sigma_j X_{\mathcal{M}^*}(\theta, \beta_i) - \Phi_{\mathcal{M}^*}(\sigma_j, \beta_i; \sigma, \beta_i)) \\
=&\Phi_{\mathcal{M}^*}(\sigma_j, \beta_i; \sigma_j, \beta_i) - \Phi_{\mathcal{M}^*}(\sigma_j, \beta_{i-1}; \sigma_j, \beta_{i-1}) \geq 0,
\end{aligned}$$

where the inequality follows from Lemma 9. □